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FACULTY OF APPLIED SCIENCES AND TECHNOLOGY

SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

PROJECT TITLE:

DEVELOPMENT OF A WEB-BASED LABORATORY INFORMATION MANAGEMENT SYSTEM (LIMS): A CASE OF YALA SUB-COUNTY HOSPITAL

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SUBMISSION DATE: APRIL 2026

DECLARATION

I hereby declare that this is my original work and has not been presented previously in another institution for any academic certificate, diploma or degree.

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DEDICATION

I dedicate this work to my family for their unwavering financial, moral, and motivational support throughout every stage of this project. To my supervisors, Dr. Edwin Ngwawe and Mr. Peter Maina Kariuki, whose guidance shaped this work. To the staff of Yala Sub-County Hospital for their cooperation. And to the Almighty God for His divine protection and provision of good health and knowledge throughout this journey.

ACKNOWLEDGEMENT

First and foremost, I would like to thank the Almighty God for great health, wisdom, and knowledge throughout the duration of this project.

I sincerely thank my supervisors, Dr. Edwin Ngwawe and Mr. Peter Maina Kariuki, for their continuous support, constructive critique, and guidance during the development and documentation of this project.

My gratitude goes to the management and staff of Yala Sub-County Hospital — particularly the Laboratory In-Charge and Medical Superintendent — for granting access to the facility and cooperating during data collection, system analysis, and User Acceptance Testing.

Lastly, I thank my family and classmates for their invaluable support throughout the research and development period.

ABSTRACT

Medical record management at sub-county level public health facilities in Kenya remains heavily paper-based, resulting in significant inefficiencies in laboratory service delivery. At Yala Sub-County Hospital, the existing manual workflow — characterised by handwritten MOH 204 request forms and MOH 240 laboratory registers — causes turnaround time delays, transcription errors, lost result slips, and poor data traceability, all of which directly impair patient care.

This project presents the design, development, and deployment of a web-based Laboratory Information Management System (LIMS) for Yala Sub-County Hospital. The system was built using PHP (procedural architecture), MySQL as the relational database engine, and Bootstrap 5 for a responsive frontend interface, deployed over the hospital's Local Area Network (LAN) via XAMPP.

The system implements comprehensive core operational modules: Patient Registration with unique OPD numbering and A5 printable routing slips featuring Code128 barcodes, Electronic Test Ordering (CPOE) with clinical allergy warnings, Laboratory Testing Workbench with automated reference range validation and critical panic value alerting, Inpatient Blood Bank crossmatching, Specimen Rejection QA (ISO 15189), Cashless Revenue Collection via Kenya Government eCitizen (Paybill 222222) with dual-emblem A5 receipting, and an Administrative Dashboard with automated MOH 706 statistical reporting. Security features include bcrypt password hashing, CSRF token protection, MySQLi prepared statements for SQL injection prevention, strict Role-Based Access Control (RBAC) across four user roles, and an immutable audit trail.

User Acceptance Testing (UAT) was conducted with 8 hospital staff members representing all user roles. The overall average satisfaction score was 4.6 out of 5.0, with all participants expressing a clear preference for the digital system over the existing manual process. The project demonstrates that a cost-effective open-source web system can successfully bridge the digital divide in resource-constrained public health facilities.

Keywords: Laboratory Information Management System, LIMS, PHP, MySQL, Role-Based Access Control, Yala Sub-County Hospital, Kenya, M-Pesa, Health Informatics.

TABLE OF CONTENTS

DECLARATION 2
DEDICATION 3
ACKNOWLEDGEMENT 4
ABSTRACT 5
TABLE OF CONTENTS 6
LIST OF FIGURES 9
LIST OF TABLES 10
CHAPTER 1: INTRODUCTION 11
1.1 Introduction 11
1.2 Background of the Study 11
1.3 Problem Statement 11
1.4 Objectives 11
1.4.1 General Objective 11
1.4.2 Specific Objectives 11
1.5 Justification 13
1.6 Scope of the Study 13
1.7 Limitations of the Proposed System 13
1.8 Project Risk and Mitigation 13
1.9 Project Schedule 13
1.10 Budget and Resources 14
CHAPTER 2: LITERATURE REVIEW 16
2.1 Reviewed Similar Systems 16
2.1.1 The Manual Paper-Based System (MOH Standard) 16
2.1.2 OpenMRS (Open Medical Record System) 16
2.1.3 Commercial LIMS (Proprietary Systems) 16
2.2 Tools and Methodologies Used in Reviewed Systems 16
2.3 Gaps in Existing Systems and Proposed Solution 16
CHAPTER 3: METHODOLOGY 17
3.1 Methodology and Tools 17
3.1.1 Methodology 17
3.1.1.1 Justification for Agile 17
3.1.1.2 Agile Sprints 17
3.1.2 System Design Tools 18
3.2 Source of Data 18
3.3 Data Collection Methods 18
3.3.1 Observation 18
3.3.2 Questionnaires 18
3.3.3 Interviews 18

3.4 Resources Required 18
CHAPTER 4: SYSTEM ANALYSIS AND REQUIREMENT MODELLING 19
4.1 Introduction 19
4.2 Feasibility Study 19
4.2.1 Technical Feasibility 19
4.2.2 Economic Feasibility 19
4.2.3 Operational Feasibility 19
4.2.4 Schedule Feasibility 20
4.3 System Requirements 20
4.3.1 Functional Requirements 20
4.3.2 Non-Functional Requirements 20
4.4 System Analysis Tools 20
4.4.1 Flowchart of Existing System 20
CHAPTER 5: SYSTEM DESIGN 22
5.1 Introduction 22
5.2 Program Design Tools 22
5.2.1 Flowchart of Proposed System 22
5.2.2 Use Case Diagram 23
5.2.3 Data Flow Diagram — Level 0 (Context Diagram) 24
5.2.4 Data Flow Diagram — Level 1 25
5.3 Logical Design — Entity Relationship Diagram 25
5.4 Physical Design 26
5.4.1 Data Dictionary 26
5.4.2 Payments & Cashless Billing Table Schema 26
5.4.3 System Roles and Permissions 27
5.4.4 Database Schema 27
CHAPTER 6: SYSTEM IMPLEMENTATION 28
6.1 Coding / Environment / Debugging 28
6.1.1 Coding Tools 28
6.1.2 Development Environment 28
6.1.3 Debugging Tools 28
6.2 Module Implementation 28
6.2.1 Authentication (index.php) 28
6.2.5 Billing & Payments (billing.php) 28
6.2.6 Reports & Analytics (reports.php) 29
6.3 System Testing 30
6.4 Proposed Change-Over Technique 30
6.5 Sample Run — System Screenshots 31
CHAPTER 7: USER MANUAL — DOCUMENTATION 46
7.1 Installation Environment 46
7.2 Installation Requirements 46

7.3 Installation Procedures 46
7.4 User Instructions 46
7.4.2 Register a Patient 46
7.4.3 Order Laboratory Tests 46
7.4.4 Enter Test Results 46
7.4.5 Record Payment (Cashier) 46
7.5 System Conversion Method 47
7.6 User Training 47
7.7 File Conversions 47
CHAPTER 8: LIMITATIONS, CHALLENGES, CONCLUSIONS AND RECOMMENDATIONS 48
8.1 Limitations 48
8.2 Challenges 48
8.3 Degree of Success 48
8.4 Learning Experience 50
8.5 Recommendations 50
8.6 Conclusion 50
REFERENCES 51
APPENDIX 52
Appendix A: User Requirements Questionnaire 52
Appendix B: Print Codes 53

LIST OF FIGURES

Figure 1: Project Gantt Chart 15
Figure 2: Budget and Resources — Proportional Breakdown 17
Figure 3: Agile Methodology Diagram 24
Figure 4: Hospital LAN Deployment Topology 26
Figure 5: Three-Tier System Architecture 28
Figure 6: Flowchart of Existing Manual System 38
Figure 7: Flowchart of Proposed Digital System 40
Figure 8: Use Case Diagram 41
Figure 9: Level 0 Data Flow Diagram (Context) 42
Figure 10: Level 1 Data Flow Diagram 43
Figure 11: Login / Sign In Screen 54
Figure 12: Dashboard — Key Performance Indicators (KPIs) 55
Figure 13: Dashboard — Core Diagnostic & Clinical Modules 56
Figure 14: Dashboard — Quality Control & Administration Modules 57
Figure 15: Register New Patient Screen 58
Figure 16: Register Patient — Statutory Insurance Details Section 59
Figure 17: Diagnostic Test Ordering — Doctor's Consultation Console 60
Figure 18: Laboratory Workbench — Active Worklist Queue 61
Figure 19: Results Entry Screen — Paid Diagnostic Request 62
Figure 20: Results Entry Screen — Unpaid Request (Payment Notice) 63
Figure 21: Patient Diagnostic History Screen 64
Figure 22: Patient History — Longitudinal Visit Details with Results 65
Figure 23: Lab Reagent Inventory — Stock Overview & Expiry KPIs 66
Figure 24: Lab Reagent Inventory — Stock Register & Batch Tracking 67
Figure 25: Quality Control (QC) & Cold Chain Equipment Log (ISO 15189) 68
Figure 26: Blood Bank & Transfusion — ABO/Rh Inventory & Reserve Alert 69
Figure 27: Blood Bank — Available Unit Inventory & TTI Screening Register 70
Figure 28: Blood Bank — Inpatient Compatibility Crossmatch Modal 71
Figure 29: Specimen Rejection QA — Pre-Analytical Indicators (ISO 15189) 72
Figure 30: Specimen Rejection Register & Phlebotomy Redraw Workflow 73
Figure 31: Cashier POS & Billing — eCitizen Paybill 222222 Gateway 74
Figure 32: Cashier POS — Pending Patient Settlements & System Footer 75
Figure 33: Government eCitizen Payment Modal — Paybill 222222 Clearance 76
Figure 34: eCitizen Cashless Settlement — STK Push & M-Pesa Verification 77
Figure 35: Hospital Intranet Footer — Standards & Repository Traceability 78
Figure 36: Reports & Analytics Dashboard — Epidemiological Summary 79
Figure 37: Reports — Daily Trend Chart and Top Requested Investigations 80
Figure 38: Reports — Recent Financial Transactions Register 81
Figure 39: System Administration — Manage Users Screen 82
Figure 40: Test Catalogue — Add New Investigation Modal 83
Figure 41: System Audit Log & Security Traceability Screen 84
Figure 42: Bulk Patient Import Screen (CSV Ingestion) 85
Figure 43: Database Backup & Maintenance Screen 86

LIST OF TABLES

Table 1: Budget and Resources 17
Table 2: Hardware and Software Requirements 28
Table 3: Data Dictionary — Patients Table (Key Fields) 47
Table 3b: Payments Table Schema (Actual) 47
Table 4: System Roles and Permissions Matrix 48
Table 5: Test Cases and Results 70
Table 6: UAT Feedback Summary 73

CHAPTER 1: INTRODUCTION

1.1 Introduction

The modernisation of healthcare infrastructure is a central pillar of Kenya's Vision 2030 and the National Digital Health Strategy. While investments have been made in digitising records at national referral hospitals, the digital divide between national and sub-county level facilities remains significant. Yala Sub-County Hospital operates its laboratory entirely on manual paper-based workflows, resulting in a cascade of inefficiencies that directly compromise the quality and timeliness of patient care. This project addresses this gap by developing a localised, cost-effective, web-based Laboratory Information Management System (LIMS) specifically tailored for the hospital's operational context.

1.2 Background of the Study

Yala Sub-County Hospital is a Level 4 public health facility in Gem Sub-County, Siaya County, Western Kenya. As a primary referral centre for the region, it handles a high volume of outpatients and inpatients daily, offering laboratory services from basic haematology to viral load testing. Despite its critical role, the laboratory department operates entirely manually — relying on handwritten MOH 204 request forms, MOH 240 laboratory registers, and physical result slips carried between departments by patients or messengers. This system is slow, error-prone, and impossible to audit effectively.

1.3 Problem Statement

The reliance on paper-based registers at Yala Sub-County Hospital results in significant laboratory turnaround time delays, as staff spend disproportionate time transcribing data rather than analysing samples. The manual system lacks a reliable audit trail, making it difficult to establish who authorised a result or when a sample was collected. Transcription errors from illegible handwriting on MOH forms are a frequent source of patient safety incidents. The loss of physical result slips affects approximately 12% of all requests per month. Monthly MOH report compilation takes an average of two full working days due to manual tallying. There is a critical need for a digital solution that automates these workflows while remaining affordable and sustainable for a public hospital.

1.4 Objectives

1.4.1 General Objective

To design, develop, and evaluate a web-based Laboratory Information Management System (LIMS) for Yala Sub-County Hospital to automate diagnostic workflows, ensure data accuracy, and improve clinical turnaround times.

1.4.2 Specific Objectives

- 1. To assess and model user requirements and manual workflow bottlenecks across outpatient, clinical consultation, and laboratory testing departments at Yala Sub-County Hospital.
- 2. To design the 3-tier system architecture, relational database schema, and role-based access interfaces (RBAC) for Receptionist, Medical Doctor, Laboratory Technologist, Cashier, and Administrator.
- 3. To develop and implement the core diagnostic modules: MOH 204 outpatient routing with Code128 barcodes, CPOE test ordering with allergy flags, laboratory results entry with automated reference ranges and panic alerting, inpatient blood bank safety, specimen

rejection QA (ISO 15189), and cashless revenue collection via Kenya Government eCitizen (Paybill 222222).

- 4. To test, validate, and evaluate the system's performance, security, and automated MOH 706 statistical reporting through User Acceptance Testing (UAT) with hospital staff.

-

1.5 Justification

The successful implementation of this project will benefit all stakeholders: the hospital gains reduced stationery costs and improved billing accuracy; clinicians gain instant access to patient test history; patients benefit from reduced waiting times and no lost results; administrators receive automated MOH-compliant reports; and the Ministry of Health gains more accurate disease surveillance data from Yala Sub-County.

1.6 Scope of the Study

In Scope: Design and development of a 3-tier web architecture on PHP 8.2 and MySQL 8.0; responsive Bootstrap 5 user interfaces with strict Role-Based Access Control; MOH 204 outpatient routing slips with Code128 barcodes; CPOE diagnostic ordering; laboratory workbench with automated reference range checking and panic alerting; inpatient blood bank crossmatching; specimen rejection QA (ISO 15189); cashless eCitizen Paybill 222222 payment gateway integration; automated FPDF PDF certification; and deployment on the hospital's intranet Local Area Network (LAN) via Apache/XAMPP. Out of Scope: Direct bidirectional serial interfacing (RS-232/HL7) with physical laboratory machines, live national cloud synchronization with DHIS2/KHIS servers, and physical hospital network cabling.

1.7 Limitations of the Proposed System

- Infrastructure dependency on a stable LAN and power supply — mitigated by UPS recommendations.
- English-only interface — may present a barrier to staff with limited English literacy.
- Long-term maintenance requires local IT support after initial deployment.

1.8 Project Risk and Mitigation

- System failure: Automated daily database backups and manual backup trigger for the Administrator.
- Unauthorised access: RBAC, bcrypt passwords, and CSRF protection implemented throughout.
- Staff resistance: Interface designed to mirror familiar manual workflow; early UAT engagement.
- Data loss during development: All source code version-controlled using GitHub.

1.9 Project Schedule

The project was executed over six months using the Agile methodology, from conceptualisation through deployment. The schedule is shown in the Gantt chart below.

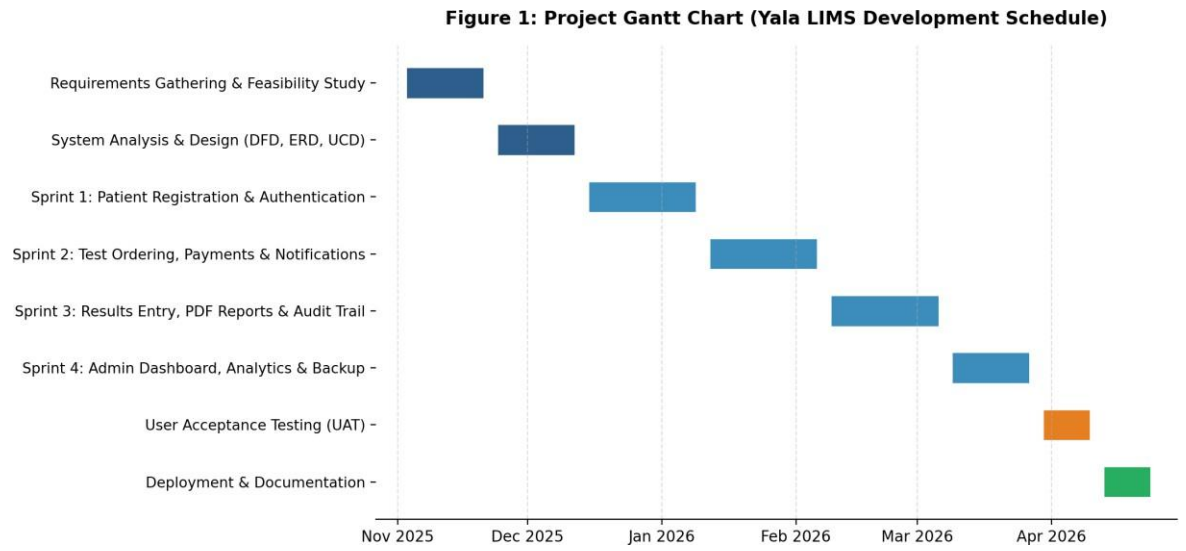


Figure 1: Project Gantt Chart

1.10 Budget and Resources

Item	Amount (KES)	Description
Laptop (Core i5, 8GB RAM)	45,000	Primary coding and deployment machine
Mobile Phone (Android)	20,000	For testing system responsiveness
Internet Access (6 months)	12,000	Research, GitHub, cloud backups
Cloud / Shared Hosting	5,000	Optional external hosting
Cloud Backup Storage	3,000	Minimum 15 GB
Antivirus Software	2,000	Kaspersky — 1-year licence
Transport & Logistics	10,000	Field visits to Yala Sub-County Hospital
Stationery & Printing	6,000	Binding of proposal and final report
Contingency	7,800	Unforeseen expenses
TOTAL	KES 110,800	

Table 1: Budget and Resources

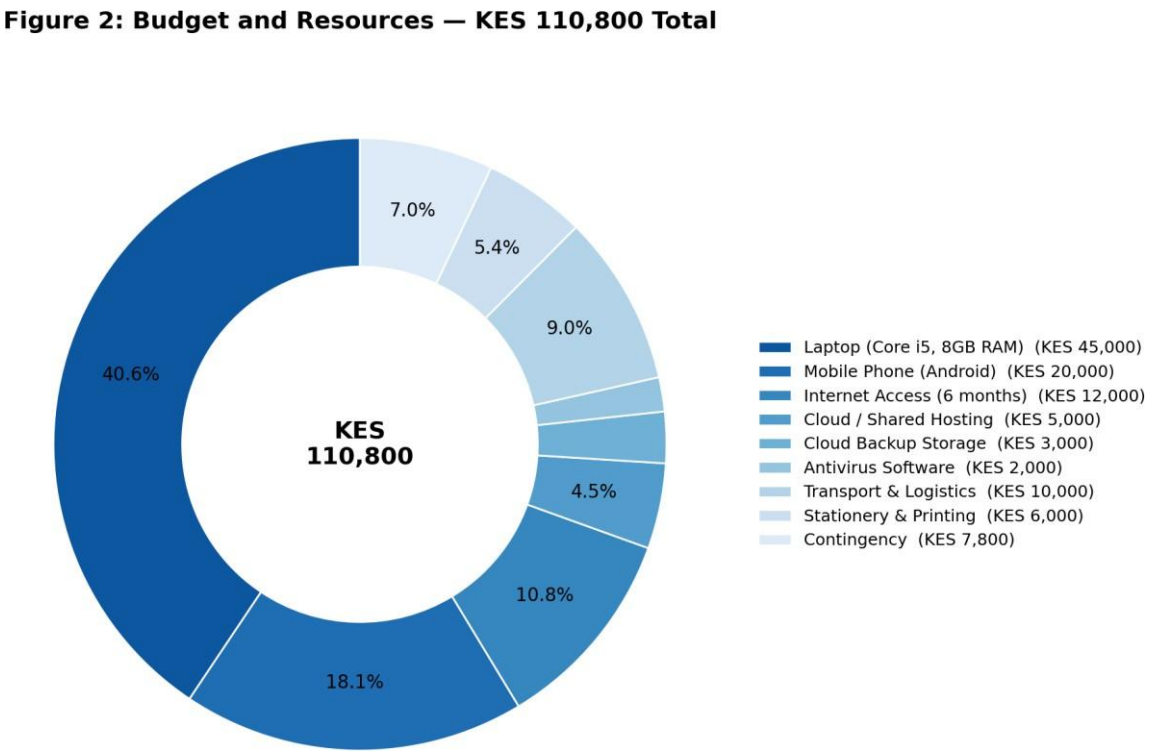


Figure 2: Budget and Resources — Proportional Breakdown

CHAPTER 2: LITERATURE REVIEW

The concept of digitising medical records to improve healthcare delivery is a subject of global interest. This chapter analyses existing systems — from the traditional manual methods at Yala Sub-County Hospital to advanced electronic platforms — to identify the gaps that the proposed system addresses.

2.1 Reviewed Similar Systems

2.1.1 The Manual Paper-Based System (MOH Standard)

This represents the current standard at the majority of Level 3 and 4 public facilities in Kenya (MOH, 2020). It relies on MOH 204 request forms and MOH 240 laboratory registers. Monthly reports are generated by manually tallying register entries. The system has been in place for decades due to its low barrier to entry and zero infrastructure requirement.

2.1.2 OpenMRS (Open Medical Record System)

An open-source EMR widely used in Kenya for HIV/AIDS programmes (AMPATH in Western Kenya). It supports longitudinal patient tracking and international data coding standards. However, its resource requirements and complexity make it unsuitable for Level 4 facilities with limited IT infrastructure.

2.1.3 Commercial LIMS (Proprietary Systems)

Systems such as LabWare LIMS, Thermo Fisher SampleManager, STARLIMS, and LabVantage provide comprehensive automation and direct analyser interfacing. Their licensing costs — often millions of shillings — make them unaffordable for county-funded public hospitals (Zahra & Patel, 2018).

2.2 Tools and Methodologies Used in Reviewed Systems

The manual system uses a transcription methodology with no logical validation. OpenMRS uses a Java/React enterprise architecture. Commercial LIMS use proprietary closed stacks. All three represent different points on the cost-complexity spectrum, leaving a gap specifically for lightweight, open-source, affordable solutions tailored to sub-county level facilities.

2.3 Gaps in Existing Systems and Proposed Solution

The manual system's gap is efficiency and accuracy. OpenMRS is too heavy and complex for Level 4 facilities. Commercial LIMS are unaffordable. The Yala LIMS addresses all three gaps: it is lightweight (runs on XAMPP over LAN), open-source (no licensing costs), and purpose-built for the specific workflows of Yala Sub-County Hospital's laboratory department.

CHAPTER 3: METHODOLOGY

3.1 Methodology and Tools

3.1.1 Methodology

The Agile Methodology was adopted for this project. Unlike the rigid Waterfall model, Agile offers an iterative and incremental approach that is well-suited to a context where requirements are discovered through close engagement with end users and where mid-development changes — such as the addition of insurance payment support — are necessary.

Figure 3: Agile Methodology — Iterative Sprint Cycle Used in Yala LIMS Development

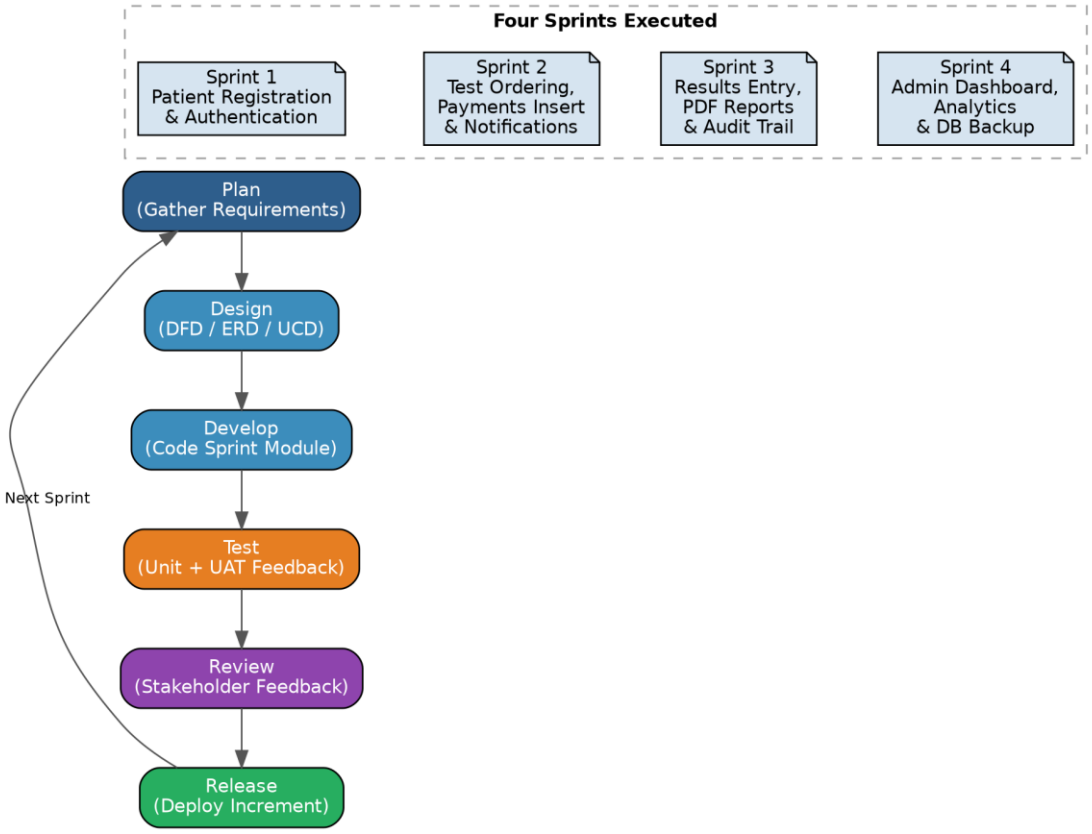


Figure 3: Agile Methodology

3.1.1.1 Justification for Agile

- Accommodates changing requirements at any stage without derailing the project.
- Delivers working modules early — each sprint produced a testable component for UAT feedback.
- Risk is managed in small iterations — a failing module can be fixed without breaking completed ones.
- Mirrors hospital staff workflows closely through continuous feedback integration.

3.1.1.2 Agile Sprints

- Sprint 1: Patient Registration and Authentication Module
- Sprint 2: Test Ordering, Payments Row Insert, and Notifications Module
- Sprint 3: Results Entry, PDF Report Generation, and Audit Trail Module
- Sprint 4: Administrative Dashboard, Reports Analytics, and Database Backup Module

3.1.2 System Design Tools

- Flowcharts: Sequential process flow for each module — drawn in Lucidchart.
- Use Case Diagrams: User-system interaction for all four roles — drawn in Visual Paradigm Online.
- Data Flow Diagrams (DFD): Level 0 and Level 1 — drawn in Lucidchart.
- Entity-Relationship Diagram (ERD): Relational database schema — drawn in dbdiagram.io.

3.2 Source of Data

Primary data was collected through field observation visits at Yala Sub-County Hospital, key informant interviews with the Lab In-Charge and Medical Superintendent, and structured questionnaires administered to 10 staff members. Secondary data was drawn from MOH standard operating procedures, open-source LIMS documentation, and academic literature on health informatics.

3.3 Data Collection Methods

3.3.1 Observation

A fly-on-the-wall approach was used during two field visits. Specific metrics recorded included average time per paper request, frequency of lost request forms, and daily test volume per technician.

3.3.2 Questionnaires

Structured questionnaires with closed and open-ended questions provided anonymity for honest staff feedback on the manual system's shortcomings.

3.3.3 Interviews

One-on-one interviews with the Lab In-Charge and Medical Superintendent provided detailed functional requirements that could not be captured by questionnaire alone.

3.4 Resources Required

Category	Resource	Spec	Unit KES	Total KES
Hardware	Development Laptop	Core i5, 8GB, 500GB SSD	90,000	90,000
	Android Mobile Device	Standard Android	20,000	20,000
	UPS / Power Backup	650VA	8,000	8,000
Software	Windows 10/11	64-bit	18,000	18,000
	VS Code, XAMPP, MySQL, GitHub	Free / Open source	0	0
	Kaspersky Antivirus	1-Year	3,000	3,000
Internet	Internet Subscription	4 Months	4,000/mo	16,000
TOTAL				155,000

Table 2: Hardware and Software Requirements

CHAPTER 4: SYSTEM ANALYSIS AND REQUIREMENT MODELLING

4.1 Introduction

This chapter presents the structural analysis of the existing manual system at Yala Sub-County Hospital and documents the detailed requirements gathered to guide the design of the proposed LIMS. It covers the feasibility study, functional and non-functional requirements, and the system design tools used to model the application's processes and data structures.

4.2 Feasibility Study

4.2.1 Technical Feasibility

The hospital has an existing Local Area Network (LAN) connecting reception, consultation rooms, and the laboratory. Workstation computers are available at reception and laboratory stations. The designated server machine meets the minimum specifications for XAMPP. No additional hardware investment is required for basic deployment. The diagram below shows how the LIMS server, network switch, and role-based workstations are physically deployed across the hospital LAN, with UPS backup protecting the server against power interruptions.

Figure 33: Hospital LAN Deployment Topology — Yala Sub-County Hospital

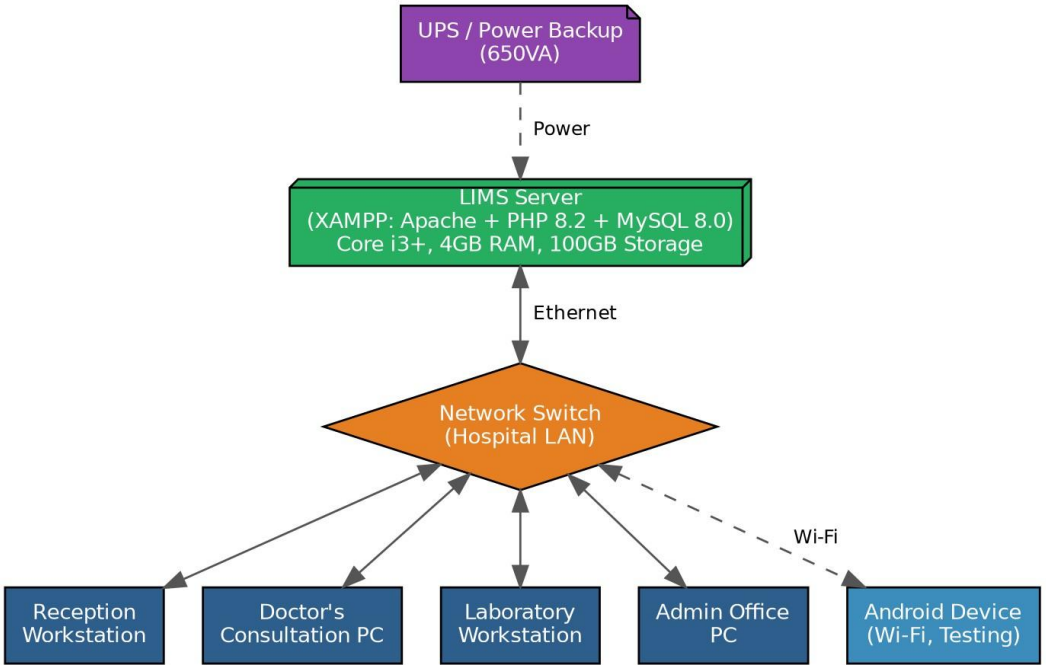


Figure 33: Hospital LAN Deployment Topology

4.2.2 Economic Feasibility

The total development cost of KES 110,800 compares favourably against the annual stationery cost and the labour cost of manual monthly MOH report compilation. The improved billing accuracy from the digital payments module is expected to recover previously un-billed laboratory services.

4.2.3 Operational Feasibility

The proposed system does not alter the fundamental workflow — test ordering, sample processing, result reporting, and billing remain the same sequential steps. The interface is designed to mirror the familiar paper-based workflow. UAT results confirmed all staff rated ease of use at 4.0 or above out of 5.0.

4.2.4 Schedule Feasibility

The project was completed within the academic calendar period through four Agile sprints across the development semester.

4.3 System Requirements

4.3.1 Functional Requirements

Patient Registration: Allow registration with unique OPD number, full name, demographics, and optional insurance details. Support bulk CSV import for existing records.

Test Ordering (CPOE): Allow Doctors to electronically select tests from a categorised catalogue, automatically calculate total cost, create a lab request, and notify the Lab Technician.

Lab Queue & Sample Tracking: Display pending/in-progress requests with status badges (Pending, In Progress, Completed, Overdue, Unpaid).

Results Entry: Allow Lab Technicians to enter result values and remarks per test, save results, update request status to Completed, and notify the ordering Doctor.

Billing & Payments: Allow Receptionists to record payments via M-Pesa, Cash, Insurance, or Waiver. Support insurance claim details and patient co-pay recording.

Administrative Dashboard: Display KPI cards, 7-day trend charts, top tests ranking, doctor activity, and recent transactions.

Audit Trail: Log all system actions with user identity, action type, affected record, and timestamp.

4.3.2 Non-Functional Requirements

Usability: Responsive Bootstrap 5 interface accessible on desktop and mobile browsers.

Performance: All pages load within 3 seconds on the hospital LAN.

Security: Bcrypt passwords, CSRF protection, MySQLi prepared statements, RBAC.

Reliability: Automated daily database backups; manual backup trigger for Admin.

Availability: System available 24/7 on the hospital LAN without internet dependency.

4.4 System Analysis Tools

4.4.1 Flowchart of Existing System

The flowchart below illustrates the current three-stage manual workflow: paper MOH 204 form → MOH 240 lab register transcription → physical result slip return to consultation room.

Figure 4: Flowchart of the Existing Manual System at Yala Sub-County Hospital

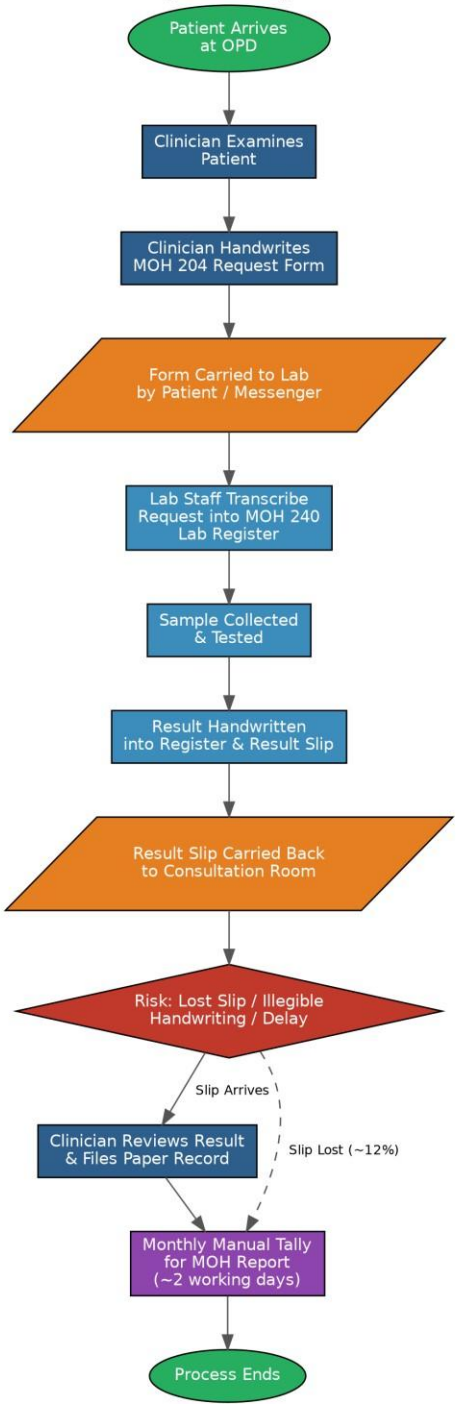


Figure 4: Flowchart of Existing Manual System

CHAPTER 5: SYSTEM DESIGN

5.1 Introduction

This chapter describes the complete design of the Yala LIMS, including the proposed system flowchart, use case diagram, data flow diagrams, entity-relationship diagram, data dictionary, database schema, and screen designs. The system follows a three-tier client-server architecture: presentation layer (Bootstrap 5), application layer (PHP 8.2 on Apache via XAMPP), and data layer (MySQL 8.0).

Figure 32: Three-Tier System Architecture — Yala LIMS

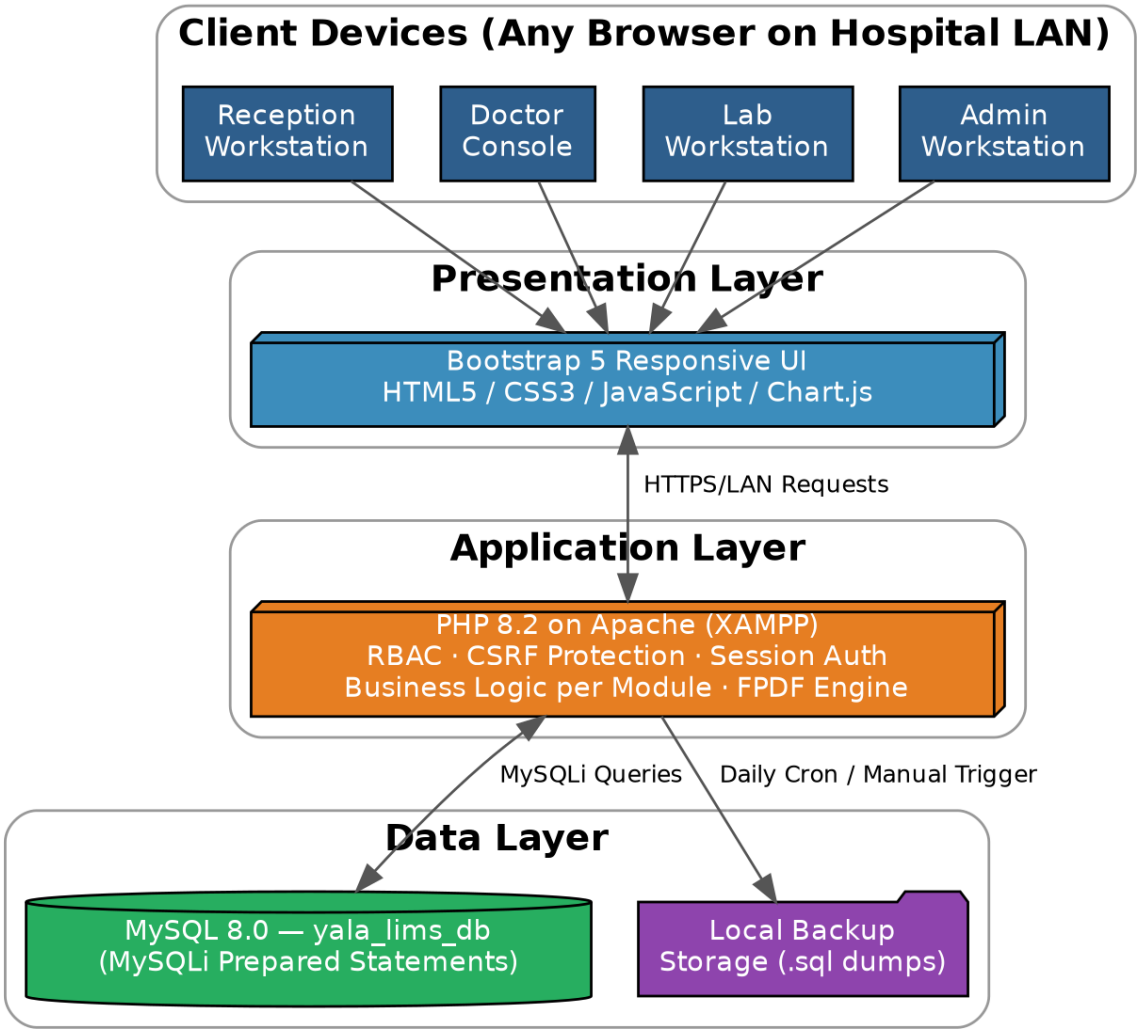


Figure 32: Three-Tier System Architecture

5.2 Program Design Tools

5.2.1 Flowchart of Proposed System

The proposed system flowchart shows the sequential digital workflow from login through patient registration, test ordering, lab processing, results entry, billing, and report generation.

Figure 5: Flowchart of the Proposed Yala LIMS Digital Workflow

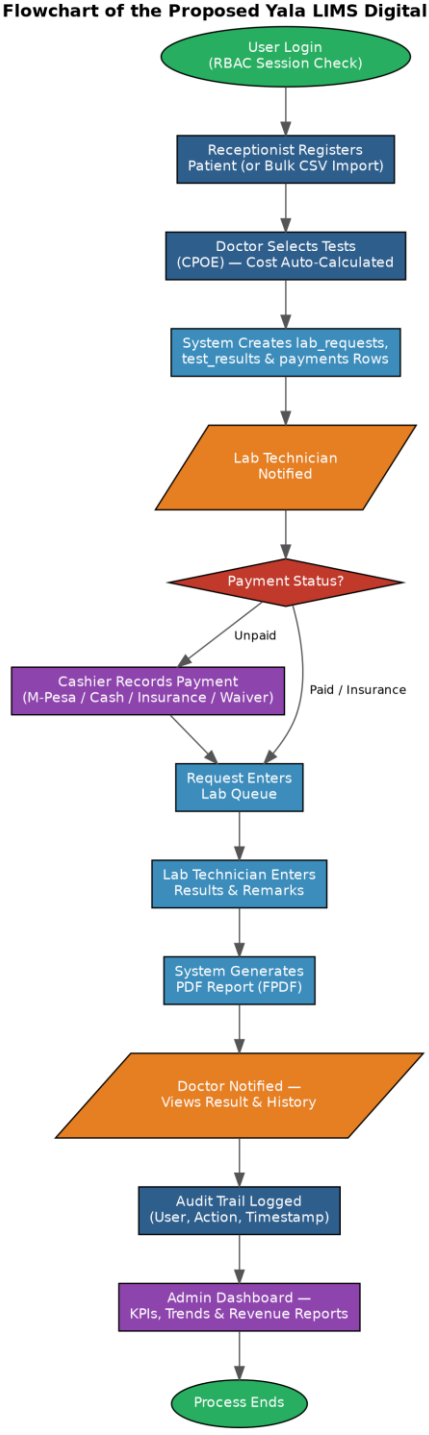


Figure 5: Flowchart of Proposed System

5.2.2 Use Case Diagram

The use case diagram shows interactions between four system actors — Receptionist, Doctor, Lab Technician, and Administrator — and the system modules available to each based on their RBAC role.

Figure 6: Use Case Diagram — Yala LIMS

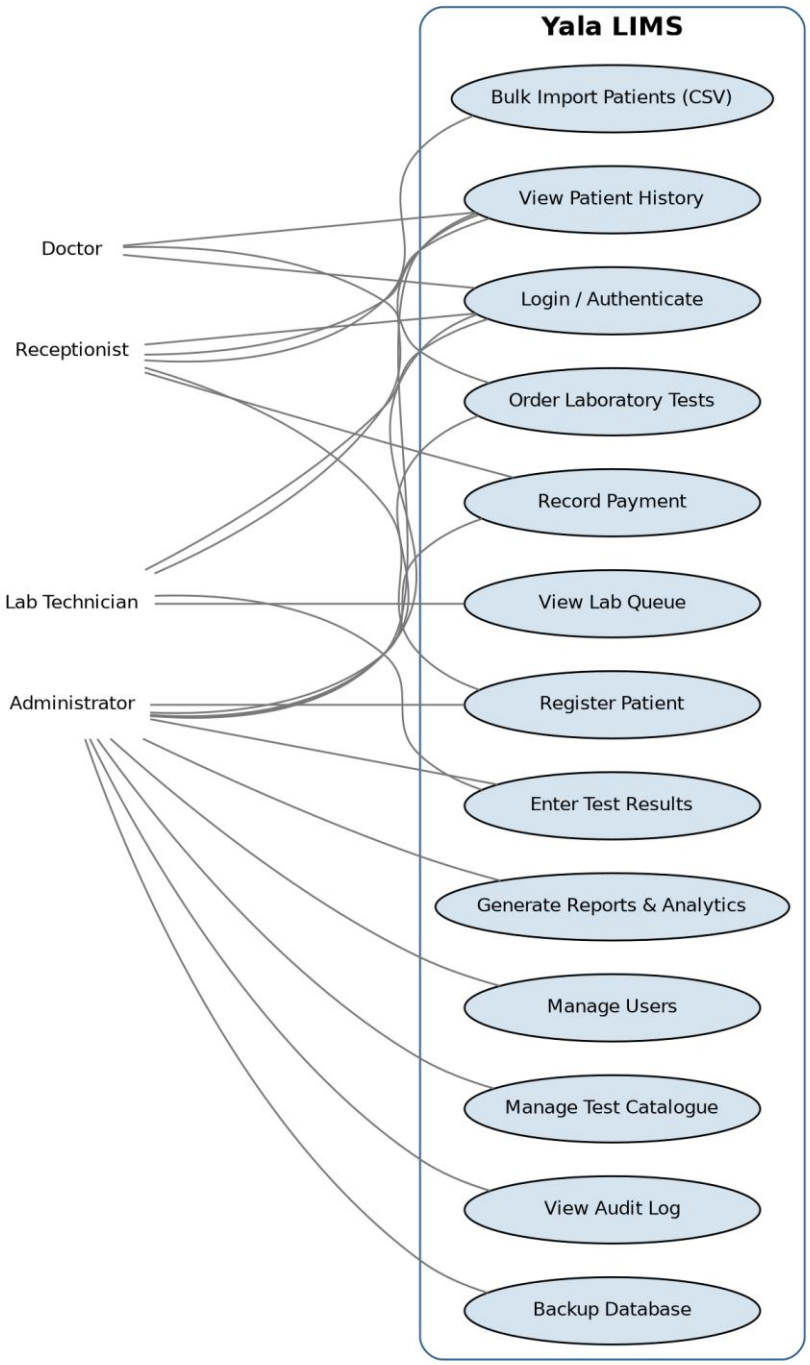


Figure 6: Use Case Diagram

5.2.3 Data Flow Diagram — Level 0 (Context Diagram)

The Level 0 DFD shows the Yala LIMS as a single process with four external entities: Doctor, Lab Technician, Receptionist/Cashier, and System Administrator.

Figure 7: Level 0 Data Flow Diagram (Context Diagram) — Yala LIMS

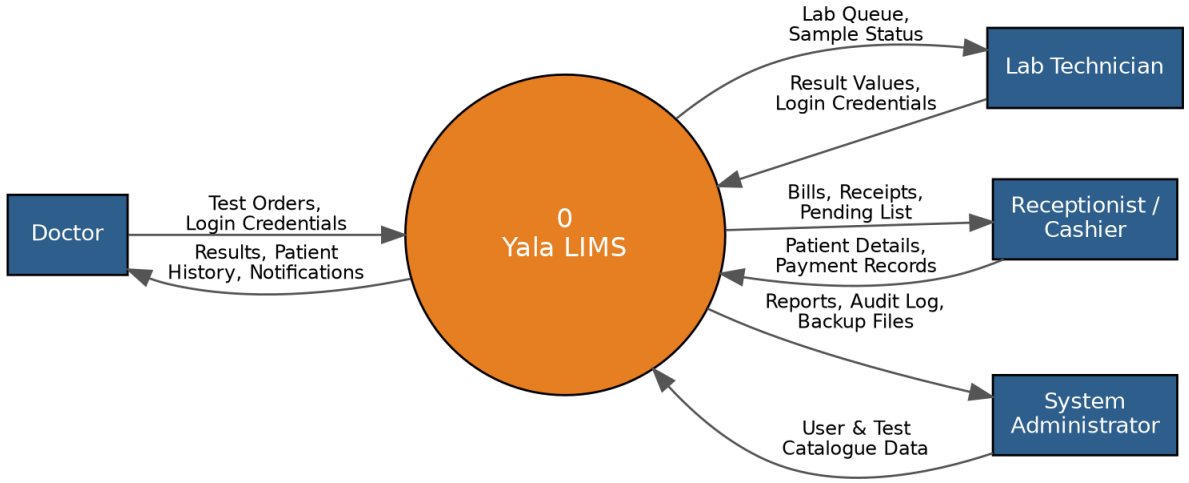


Figure 7: Level 0 Data Flow Diagram

5.2.4 Data Flow Diagram — Level 1

The Level 1 DFD decomposes the system into six sub-processes: Manage Patients, Process Test Orders, Record Test Results, Process Billing, Manage Notifications, and Generate Reports — connected to seven data stores.

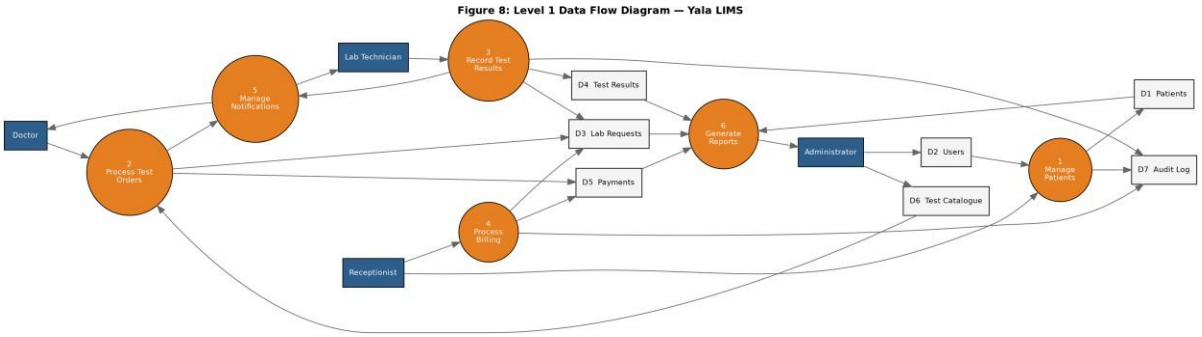


Figure 8: Level 1 Data Flow Diagram

5.3 Logical Design — Entity Relationship Diagram

The ERD models the relational structure of the yala_lims_db database.

Figure 9: Entity Relationship Diagram — yala_lims_db

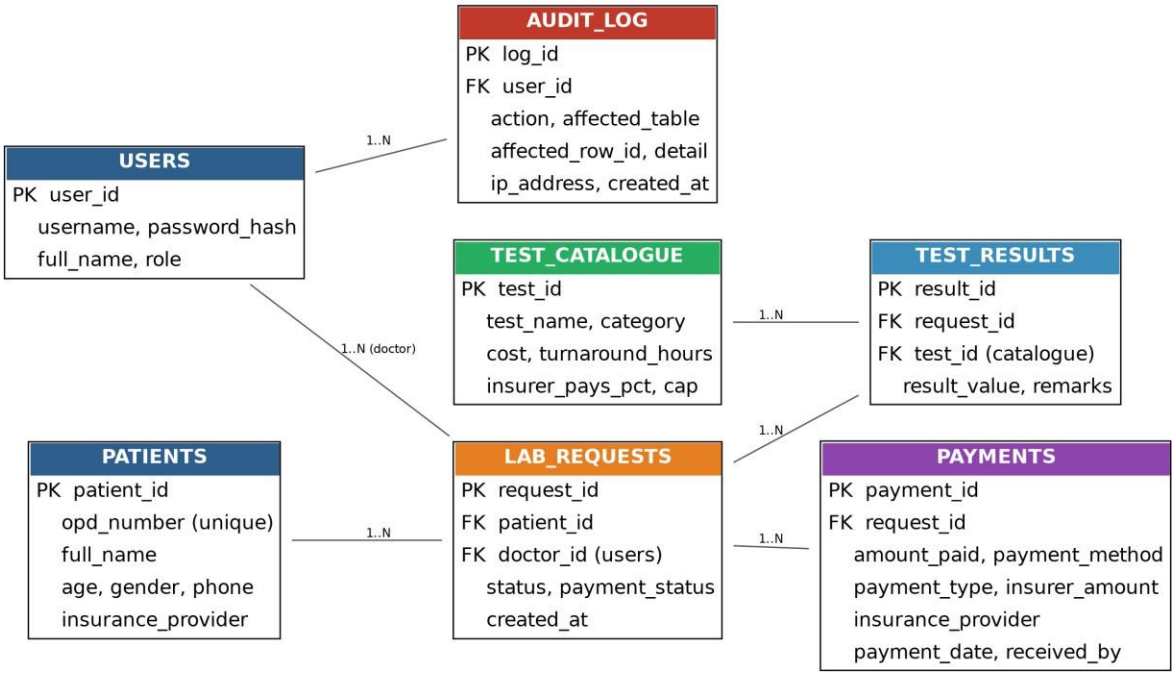


Figure 9: Entity Relationship Diagram

5.4 Physical Design

5.4.1 Data Dictionary

Field	Type	Description	Null
patient_id	INT AUTO	Unique patient identifier — primary key	NO
opd_number	VARCHAR(50)	Hospital OPD/outpatient number — unique	NO
full_name	VARCHAR(150)	Patient full name	NO
age	INT	Age in years	NO
insurance_provider	VARCHAR(100)	Insurance company (optional)	YES

Table 3: Data Dictionary — patients table (key fields)

5.4.2 Payments & Cashless Billing Table Schema

The payments table records transaction audit trails, cashless eCitizen references, statutory insurance authorizations, and cashier settlement records:

Field	Type	Description	Key
payment_id	INT AUTO	Primary key	PRI
request_id	INT	Links to lab_requests	MUL
amount_paid	DECIMAL(10,2)	Amount actually collected	
payment_method	VARCHAR(50)	'Pending' on insert; updated to mpesa/cash/insurance/waiver on settlement	
payment_type	ENUM(Cash,Insurance,Split)	Category of payment	
insurer_amount	DECIMAL(10,2)	Total bill amount (stored here since no total_cost column exists)	

insurance_provider	VARCHAR(100)	Insurer name from patient record	
payment_date	TIMESTAMP	Auto-populated on insert	
received_by	VARCHAR(150)	Name of Receptionist/Cashier who processed payment	

Table 3b: Payments Table Schema

5.4.3 System Roles and Permissions

Module	Admin	Doctor	Lab Tech	Receptionist
Dashboard	✓	✓	✓	✓
Register Patient	✓	✓	✗	✓
Order Tests (CPOE)	✓	✓	✗	✗
Enter Results	✓	✗	✓	✗
Billing/Payments	✓	✗	✗	✓
Patient History	✓	✓	✓	✓
Reports	✓	✗	✗	✗
Manage Users	✓	✗	✗	✗
Test Catalogue	✓	✗	✗	✗
Audit Log	✓	✗	✗	✗
Database Backup	✓	✗	✗	✗

Table 4: System Roles and Permissions Matrix

5.4.4 Database Schema

The screenshot below shows the physical database schema of yala_lims_db as viewed in phpMyAdmin.

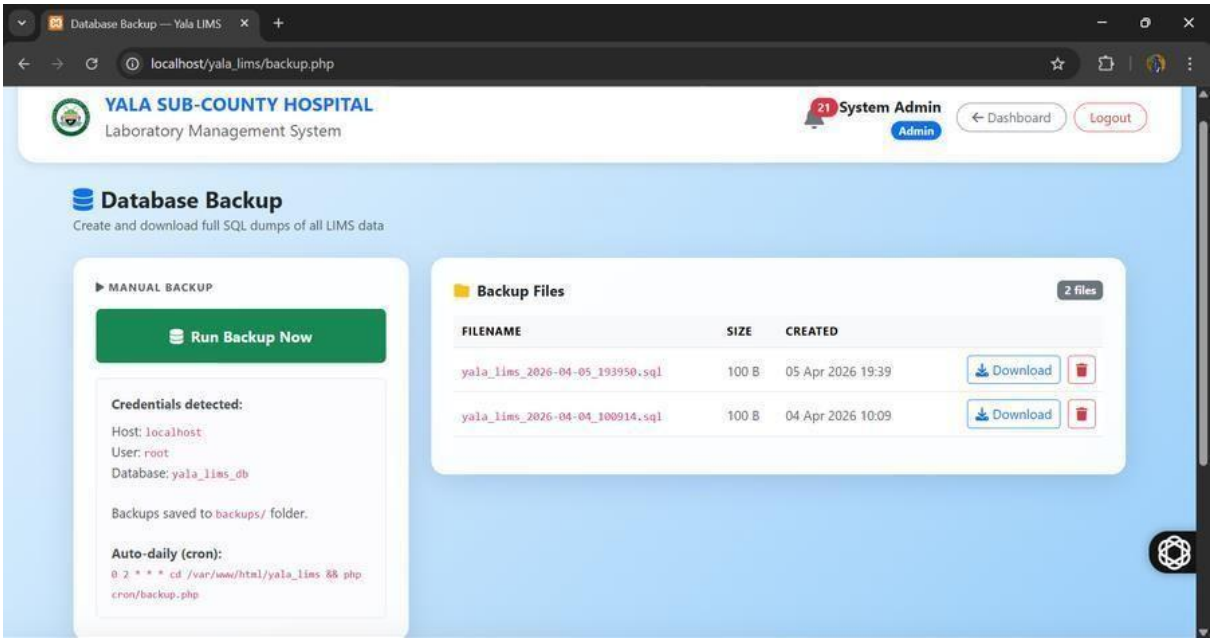


Figure 10: Database Schema (phpMyAdmin)

CHAPTER 6: SYSTEM IMPLEMENTATION

This chapter documents the actual coding environment, the implementation of each system module, testing, and sample run output screenshots from the deployed Yala LIMS system.

6.1 Coding / Environment / Debugging

6.1.1 Coding Tools

PHP 8.2 (server-side logic), HTML5/CSS3/Bootstrap 5 (frontend UI), JavaScript with Chart.js (data visualisation in Reports), MySQL 8.0 via MySQLi with prepared statements (database), FPDF 1.86 (PDF generation).

6.1.2 Development Environment

XAMPP v8.2 on Windows 11 provided Apache, PHP, and MySQL. Visual Studio Code with PHP Intelephense was used as the IDE. All code was version-controlled using GitHub throughout development.

6.1.3 Debugging Tools

PHP error_reporting(E_ALL) during development, Chrome DevTools for frontend debugging, phpMyAdmin for database query testing, and GitHub version history for rollback capability.

6.2 Module Implementation

6.2.1 Authentication (index.php)

Login uses MySQLi prepared statements to fetch the user record by username, then verifies the submitted password against the stored bcrypt hash using password_verify(). On success, role, full name, and user ID are stored in the PHP session. Every protected page begins with a session check.

6.2.2 Patient Registration (add_patient.php)

The registration form captures OPD number, full name, demographics, and optional insurance details. The system validates OPD uniqueness before inserting. Dynamic schema checks detect whether optional columns exist, enabling backward compatibility. Audit log is written after each successful registration. Bulk CSV import is supported via import_patients.php.

6.2.3 Test Ordering (request_test.php)

Doctors search for a patient by OPD number, then select required investigations from the categorised laboratory catalogue. The system records the clinical request, initialises test parameter records, and creates a pending billing transaction awaiting cashless settlement at the cashier desk. An asynchronous notification is immediately dispatched to the laboratory workbench queue upon submission.

6.2.4 Lab Queue & Results Entry (enter_results.php)

The Lab Workbench displays pending and in-progress requests with status indicators including Overdue badges for requests older than 2 hours. For insurance or paid requests the lab can proceed immediately. For unpaid cash requests, a payment-required notice with a link to the cashier is shown. Lab Technicians enter result values and remarks per test; saving updates the request status to Completed and notifies the ordering Doctor.

6.2.5 Billing & Payments (billing.php)

The Cashier module retrieves all diagnostic requests awaiting settlement. In compliance with the Public Finance Management Act (PFM Act 2012) and Kenya Gazette Notice No. 16008 of 2022, physical cash handling is strictly eliminated, and all settlements are routed to Kenya Government eCitizen Paybill 222222 with dynamic patient account identifiers. The module supports instant M-Pesa STK push prompts, manual transaction reference code verification, Social Health Authority (SHA) and Linda Mama statutory coverage, and authorized fee waivers. Upon confirmation, the request is flagged as Paid, immediately unlocking result entry on the laboratory workbench.

6.2.6 Reports & Analytics (reports.php)

The Admin-only reports module executes aggregate SQL queries over a user-specified date range to produce: new patient count, total requests, completed count, pending count, completion rate, and revenue. Chart.js renders a 7-day trend chart and a gender donut chart. A top tests table shows frequency and revenue per test.

Laboratory Reagents & Consumables Inventory (inventory.php)

The Inventory module provides real-time stock surveillance for chemical reagents, rapid diagnostic test kits, specimen collection tubes, and general laboratory consumables. Each inventory item is catalogued with its commercial batch number, lot code, assigned storage condition (such as Reagent Refrigerator 2–8°C or Chemical Cabinet), expiry date, and current quantity balance. Automated visual KPI counters instantly flag depleted reagents exceeding designated minimum reorder levels, items nearing expiry within 30 days, and expired supplies. Technicians can execute immediate stock adjustments (+10 units or one-touch dispensing) with full timestamped audit trail logging.

Quality Control (QC) & Cold Chain Equipment Log (qc_log.php)

Adhering to ISO 15189:2022 laboratory accreditation requirements, the Quality Control module maintains an automated daily audit log of all critical temperature-dependent storage units and automated analytical instruments. Hospital laboratory technologists record morning and evening temperatures for deep freezers (-20°C Fresh Frozen Plasma storage), reagent refrigerators (2–8°C), and blood bank storage units (2–6°C), as well as optical density calibration checks for clinical chemistry analyzers. The system validates recorded values against pre-configured target ranges, generating visual non-compliance warnings and documenting mandatory corrective actions.

Blood Bank & Inpatient Transfusion Safety (blood_bank.php)

The Blood Bank module manages hospital blood reserves, emergency transfusion crossmatching, and donor unit safety verification. The dashboard monitors real-time inventory across all eight ABO and Rhesus D blood groups, with automated alerts for critical deficits in universal donor O-negative units and units approaching expiration. Prior to transfusion, technologists record mandatory Transfusion-Transmissible Infection (TTI) screening results (HIV, Hepatitis B, Hepatitis C, and Syphilis). A dedicated crossmatch interface allows clinicians to link blood bag units to recipient inpatient records, document compatibility testing (absence of agglutination), assign recipient ward/bed locations, and track units marked as transfused.

Specimen Rejection & Phlebotomy QA (sample_rejection.php)

In compliance with ISO 15189:2022 clause 5.4, the Specimen Rejection module automates pre-analytical error surveillance to prevent compromised specimens from producing misleading clinical results. Laboratory staff log unsuitable specimens by selecting standardized root causes—including gross hemolysis, micro-clots in EDTA tubes, insufficient sample volume (QNS), lipemia, or inappropriate collection containers. The module continuously computes the laboratory specimen rejection rate against the national Ministry of Health benchmark of ≤ 2.0%, maintains an audit trail of rejected samples, and coordinates redraw workflows by dispatching redraw notifications directly to outpatient phlebotomy stations.

MOH 204 Outpatient Consultation Routing Slip & Barcode Generation (print_routing_slip.php)

The Consultation Routing module digitises the official Ministry of Health MOH 204 Outpatient Consultation workflow. Upon patient registration, the system automatically renders an A5-formatted clinical routing slip containing patient demographics, visit date, clinical notes, and a high-density Code128 optical barcode encoding the patient's unique OPD number. Phlebotomists and laboratory technicians scan the barcode to retrieve

patient orders instantaneously, eliminating transcription errors and guaranteeing patient sample chain of custody.

Real-Time Clinical Notifications & Panic Alerts System (notifications.php)

The notification engine provides instantaneous communication across distributed hospital LAN workstations. When a clinician submits diagnostic orders, a notification immediately alerts the laboratory workbench. When laboratory results reveal panic critical values (such as severe hypoglycemia < 2.5 mmol/L or profound anemia Hb < 7.0 g/dL), high-priority audio-visual toast alerts are broadcast to the ordering physician. The system also tracks specimen redraw requests and cashier clearance receipts in real time.

6.3 System Testing

Unit testing was conducted per module during each sprint. Integration testing verified end-to-end data flow across modules. The test cases and results are shown in the table below.

TC#	Test Description	Input	Expected Result	Status
TC01	Valid user login	Correct credentials	Redirect to Dashboard	PASS
TC02	Invalid password	Wrong password	Error message shown	PASS
TC03	Register new patient	Unique OPD, full details	Patient saved, audit logged	PASS
TC04	Duplicate OPD number	Existing OPD	Error: OPD already exists	PASS
TC05	Submit test order	Patient + 2 tests selected	Request created, lab notified, payment row inserted	PASS
TC06	Duplicate active request	Patient with active order	Error: active request exists	PASS
TC07	Enter and save results	Result values entered	Status → Completed, doctor notified	PASS
TC08	Record M-Pesa payment	M-Pesa code + amount	Payment recorded, receipt printable	PASS
TC09	Record Insurance payment	Insurer name, claim ref, copay	Payment type set to Insurance	PASS
TC10	Unauthorised module access	Doctor accessing Reports	Redirect to dashboard	PASS
TC11	CSRF token mismatch	Tampered hidden token	Request rejected, 403 error	PASS
TC12	Manual database backup	Admin clicks Backup Now	.sql file created in /backups/	PASS
TC13	Audit log on result save	Lab Tech saves results	Audit entry with user + timestamp	PASS
TC14	Bulk CSV patient import	CSV with 10 valid rows	10 patients imported, duplicates skipped	PASS

Table 5: Test Cases and Results

6.4 Proposed Change-Over Technique

A parallel conversion technique is used: the new digital system runs alongside the existing MOH manual registers for a minimum of four weeks post-deployment. This ensures data integrity verification and provides a safety fallback before the manual process is retired.

6.5 Sample Run — System Screenshots

The following screenshots show the deployed Yala LIMS system during the testing phase.

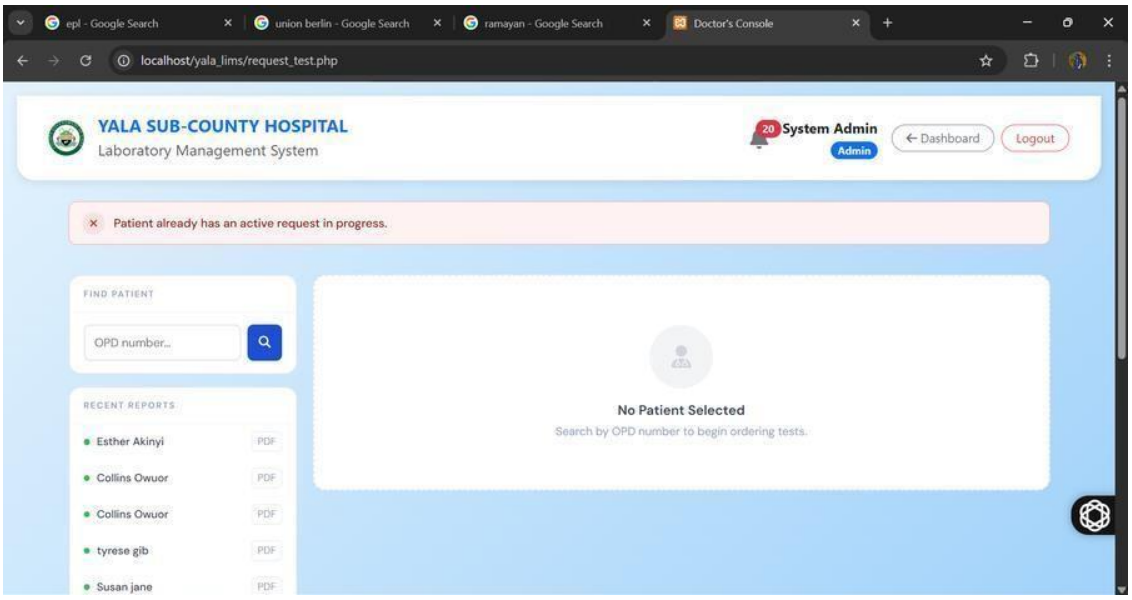


Figure 11: Login / Sign In Screen

Displays the secure authentication gateway requiring valid username, password, and active hospital role credentials. Protected with bcrypt hashing and CSRF tokens.

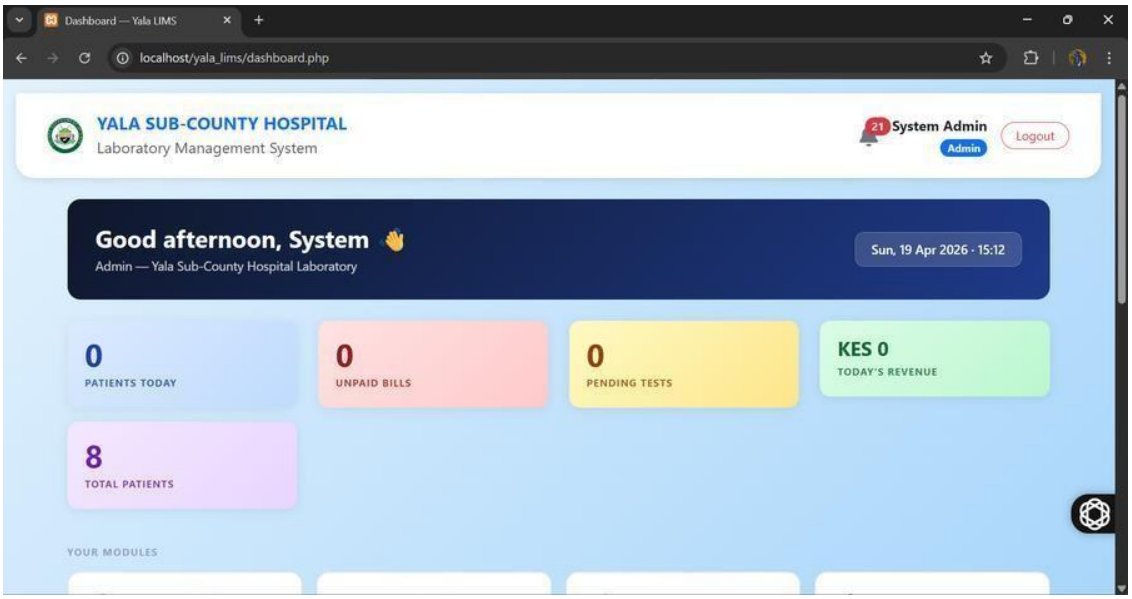


Figure 12: Dashboard — Key Performance Indicators (KPIs)

Presents high-level clinical operational indicators, including total registered patients, pending diagnostic requests, completed investigations, and revenue summary.

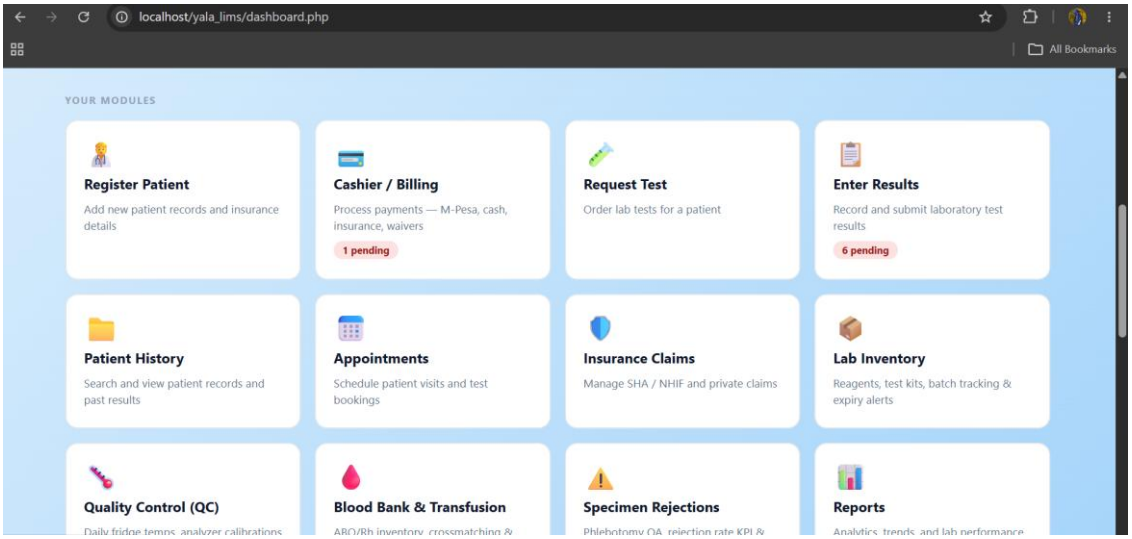


Figure 13: Dashboard — Core Operational Modules

Overview of primary diagnostic workflows: Patient Registration, Cashier / Billing, Test Ordering, Enter Results, Patient History, Appointments, Insurance Claims, and Lab Inventory.

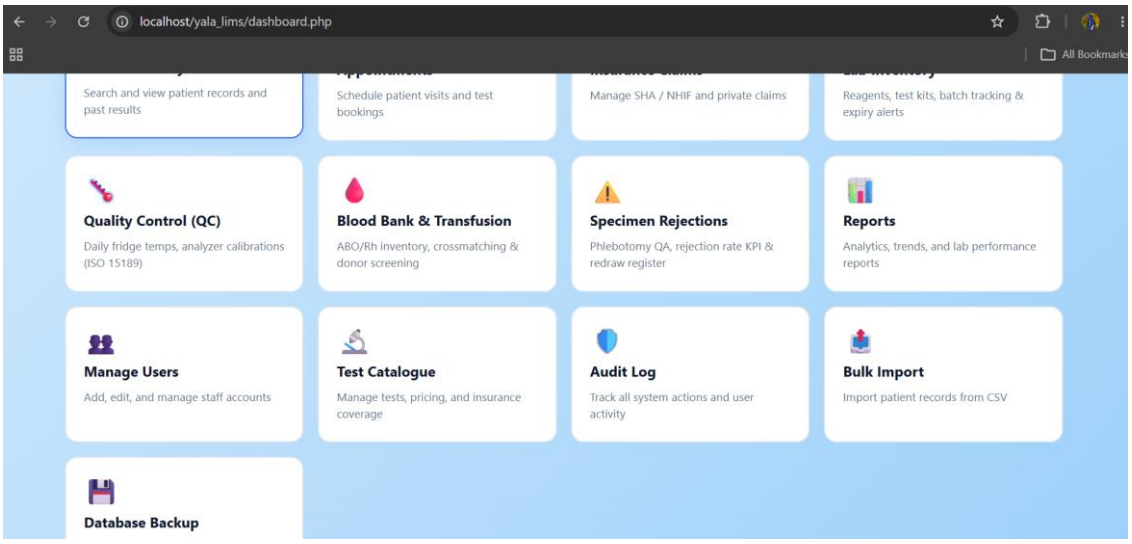


Figure 14: Dashboard — Quality Control & Administration Modules

Displays secondary clinical and administrative modules: Quality Control (QC), Blood Bank & Transfusion, Specimen Rejections, Reports & Analytics, Manage Users, Test Catalogue, Audit Log, and Backup.

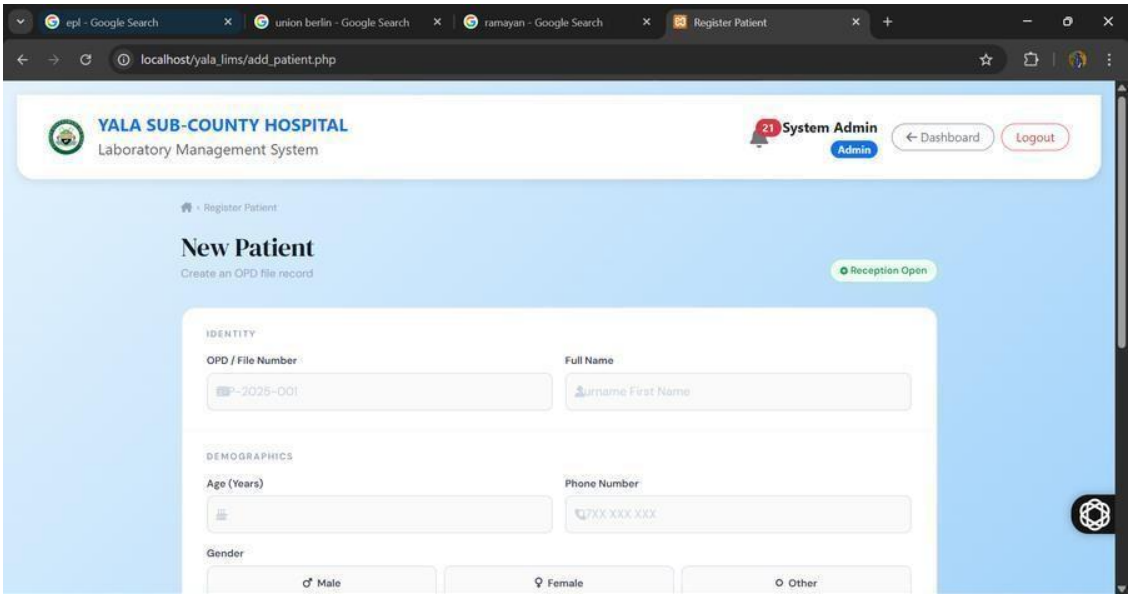


Figure 15: Register New Patient Screen

Captures comprehensive patient demographics, unique OPD identifier, national identification number, and contact details with real-time uniqueness validation.

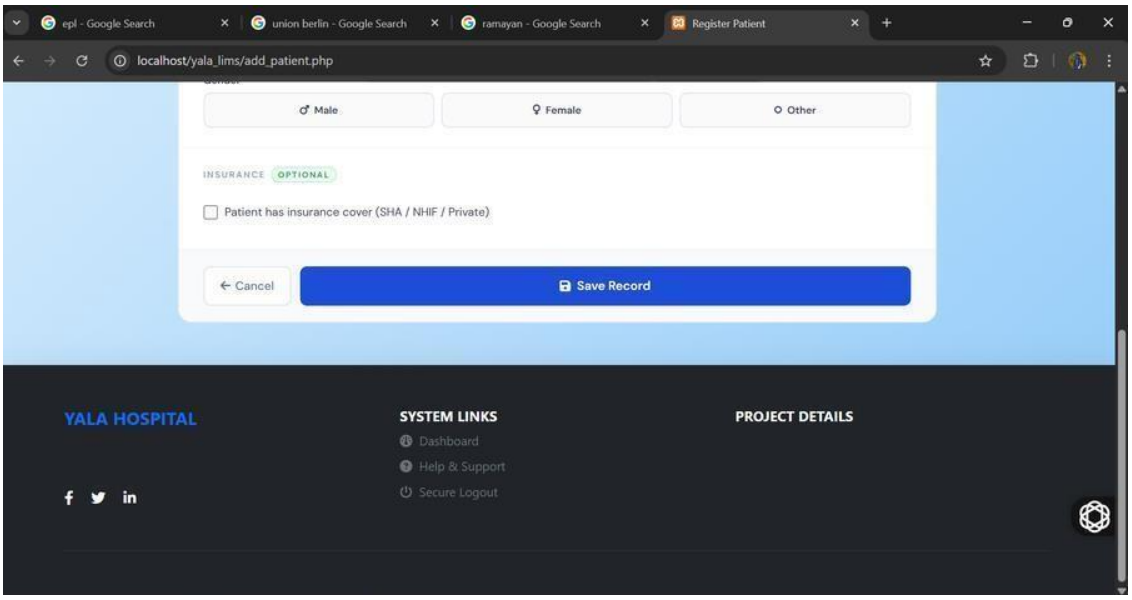


Figure 16: Register Patient — Statutory Insurance Details Section

Allows capturing patient health insurance data, including Social Health Authority (SHA / NHIF) numbers, scheme principal details, and private insurance policy numbers.

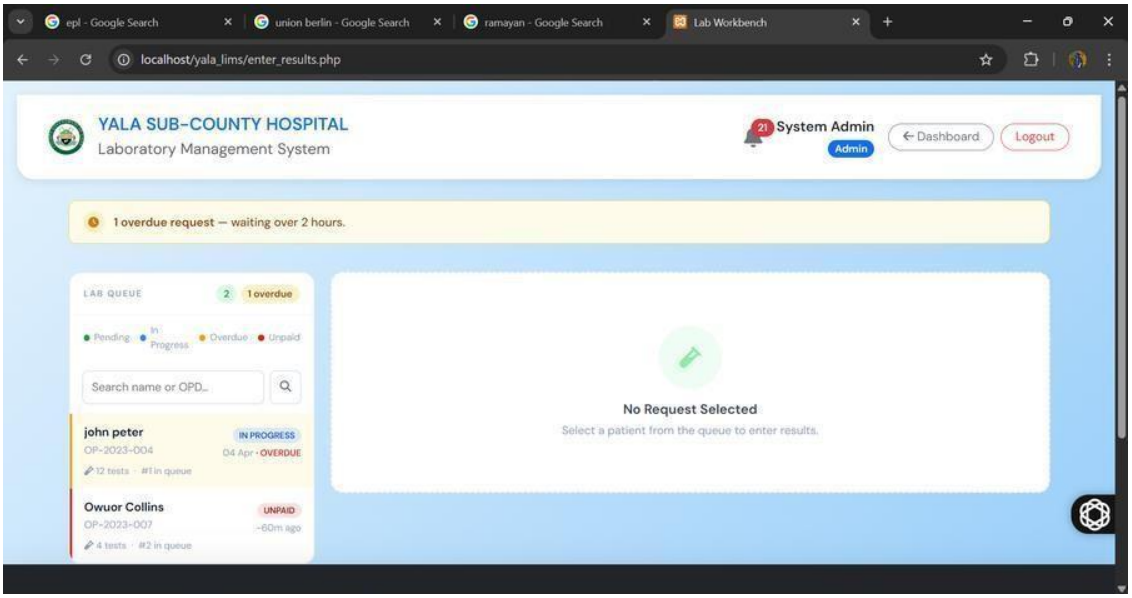


Figure 17: Diagnostic Test Ordering — Doctor's Consultation Console

Clinicians select tests across haematology, clinical chemistry, parasitology, microbiology, and serology, with automated cost calculation and allergy alerts.

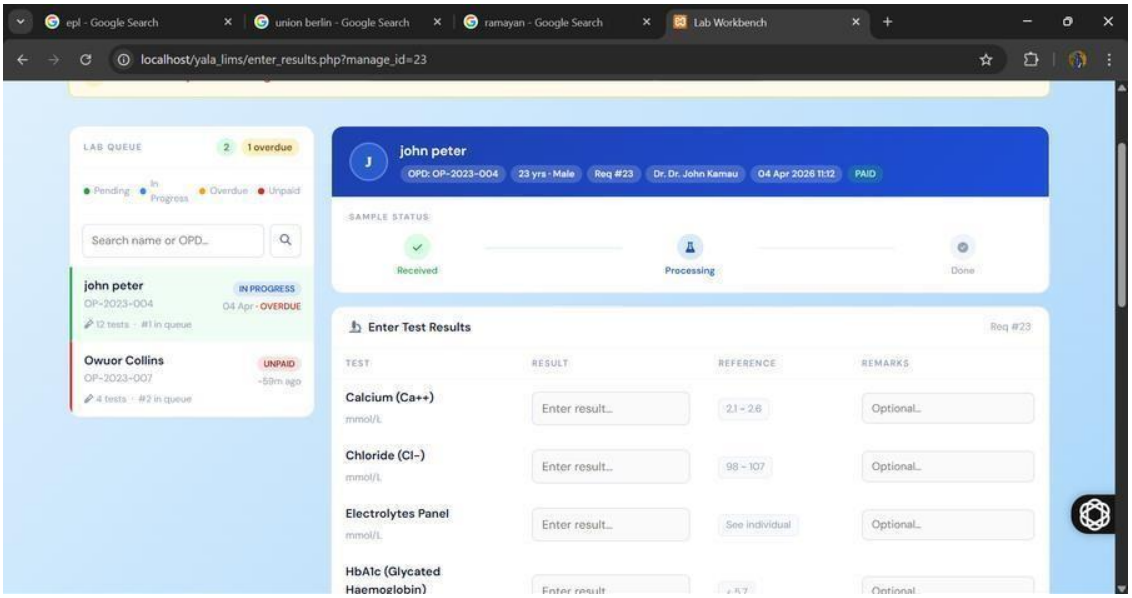


Figure 18: Laboratory Workbench — Active Worklist Queue

Displays all incoming diagnostic orders, priority indicators, specimen collection timestamps, and overdue alerts for requests waiting longer than 2 hours.

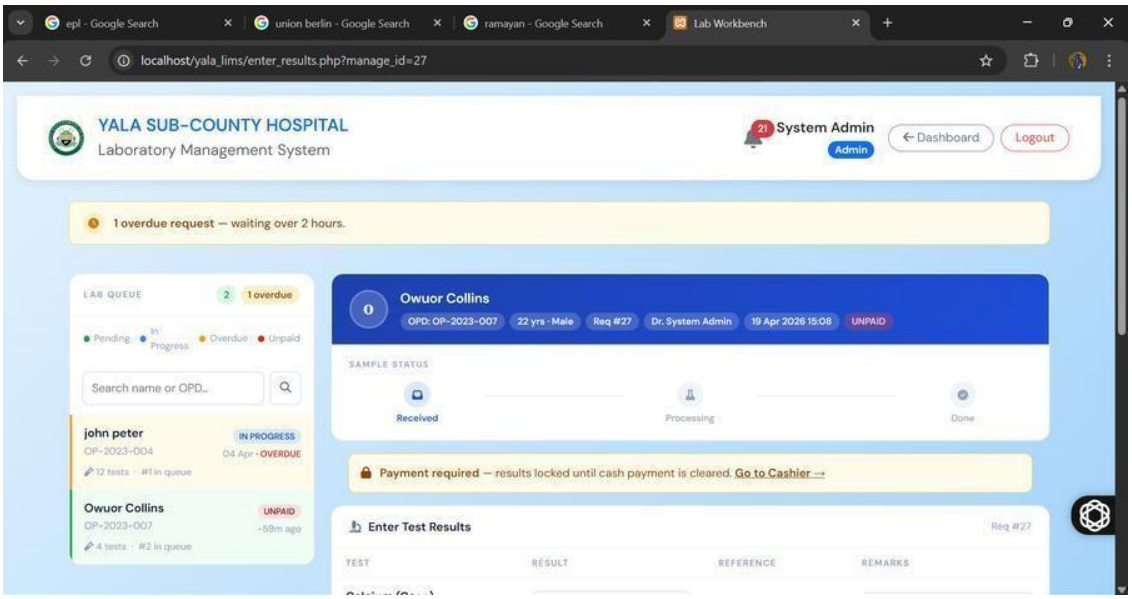


Figure 19: Results Entry Screen — Paid Diagnostic Request

Laboratory technologists record parameter findings with automated normal reference range validation, highlighting low, normal, and critical panic values.

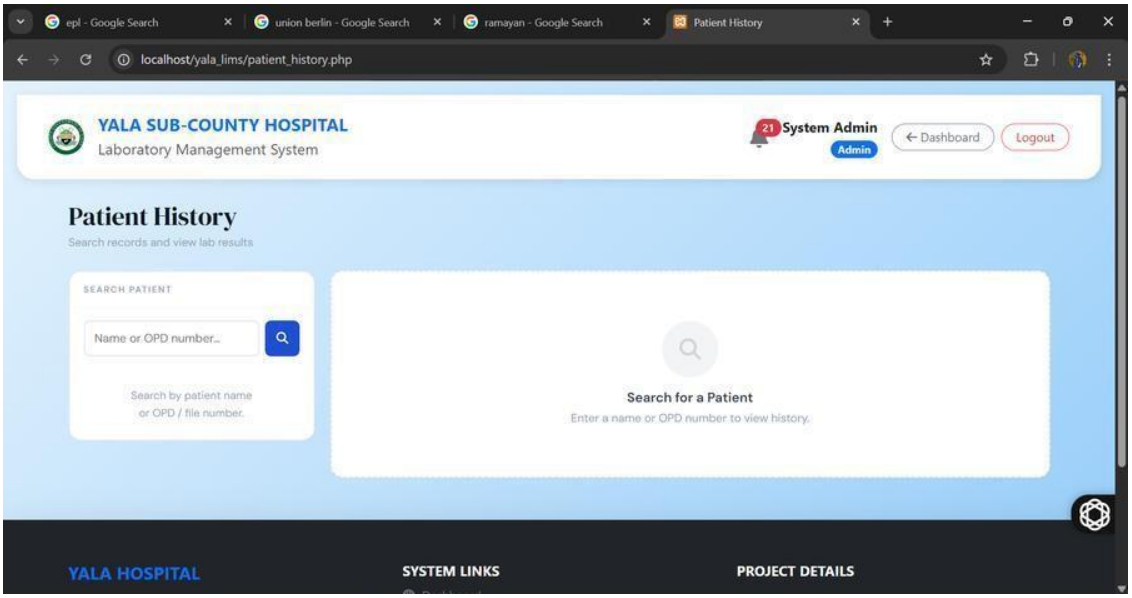


Figure 20: Results Entry Screen — Unpaid Request (Payment Clearance Notice)

Enforces revenue integrity by displaying a payment-required alert for unpaid cash patients, restricting result entry until cashier clearance is recorded.

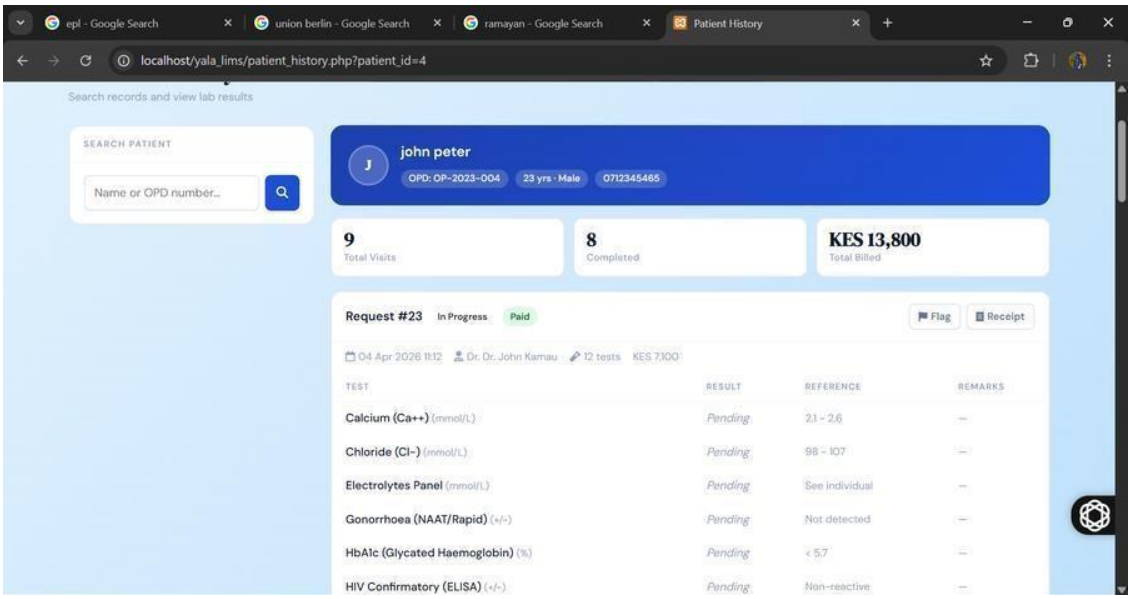


Figure 21: Patient Diagnostic History Screen

Enables fast searching of past patient records by OPD number or name, displaying longitudinal visit timelines and diagnostic summaries.

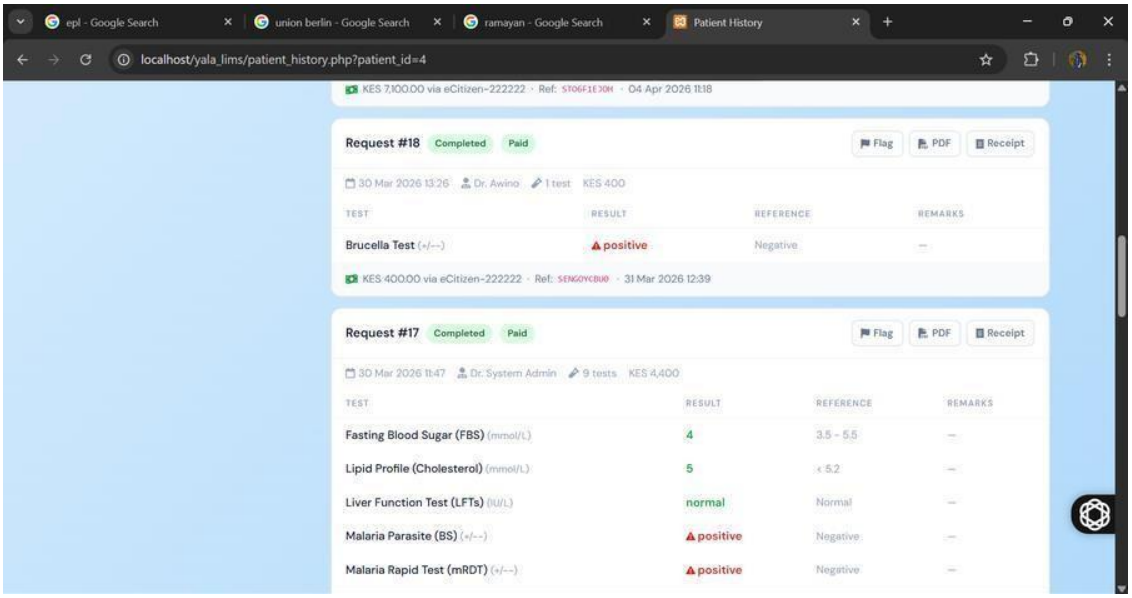


Figure 22: Patient History — Longitudinal Visit Details with Results

Detailed clinical record view displaying complete test results, reference ranges, certifying technologist, and digital PDF slip download link.

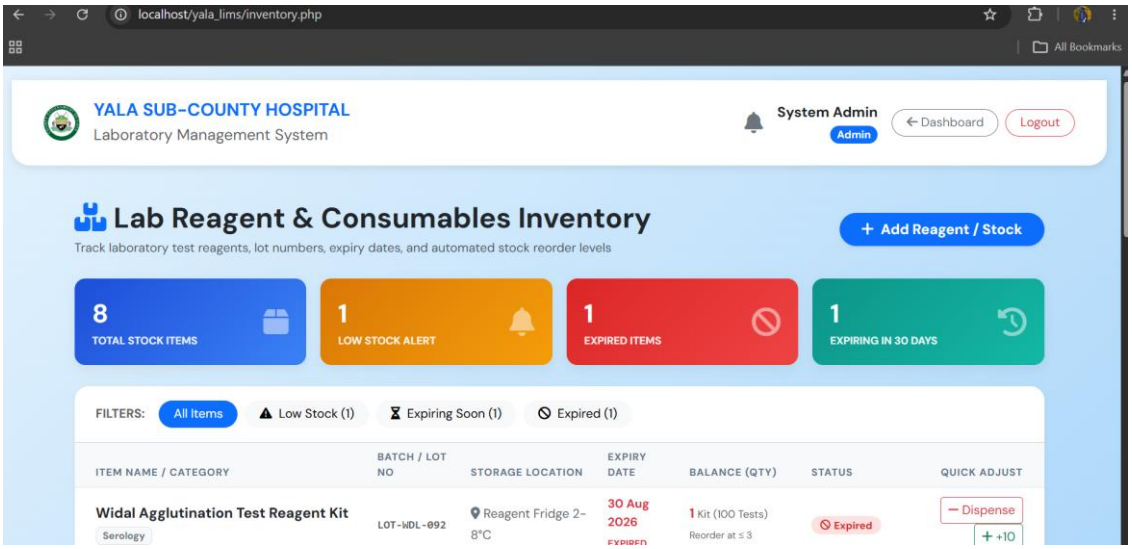


Figure 23: Lab Reagent Inventory — Stock Overview & Expiry KPIs

Real-time surveillance of laboratory reagents and consumables, displaying total stock items, low stock warnings, expired materials, and items expiring within 30 days.

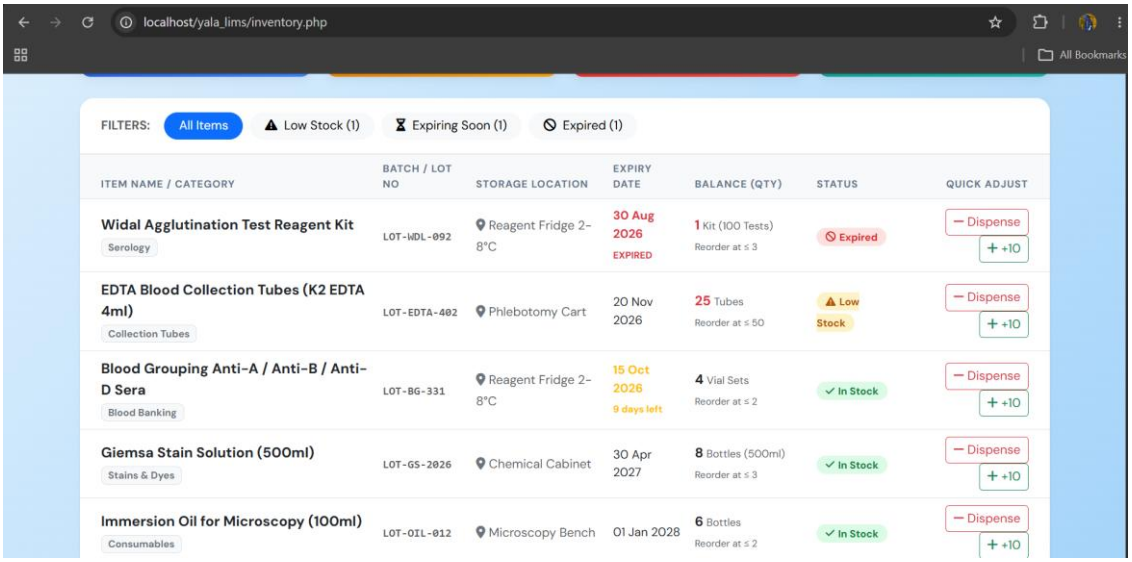


Figure 24: Lab Reagent Inventory — Stock Register & Batch Tracking

Detailed consumables registry capturing lot numbers, storage locations (e.g., Reagent Fridge 2–8°C), expiry dates, minimum reorder levels, and one-click dispense actions.

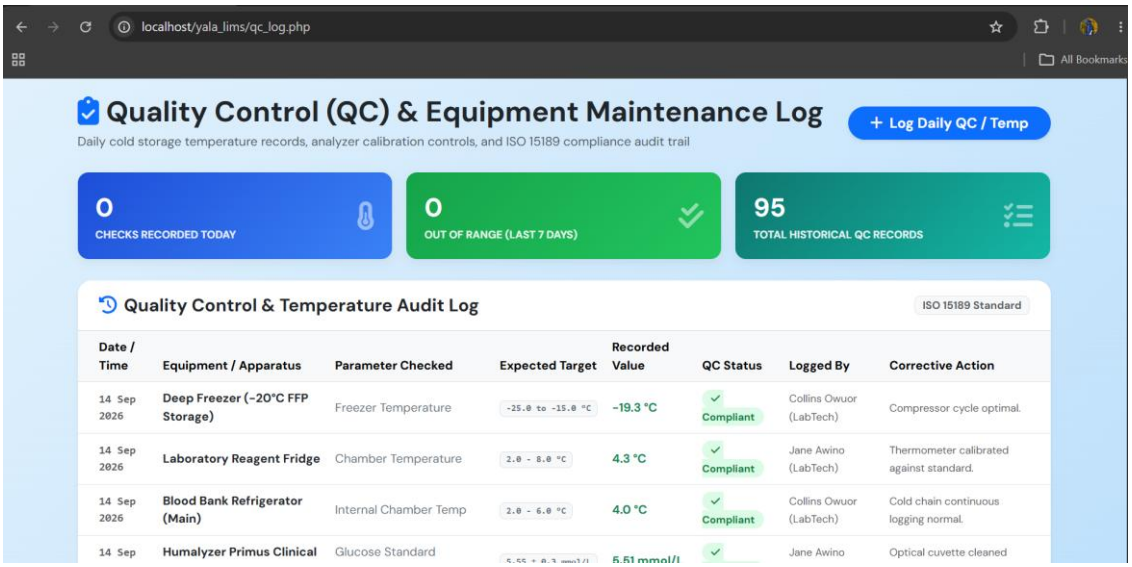


Figure 25: Quality Control (QC) & Cold Chain Equipment Log (ISO 15189)

Daily temperature audit trail for deep freezers (-20°C FFP storage), reagent refrigerators (2–8°C), blood bank storage (2–6°C), and automated analyzer calibration standards.

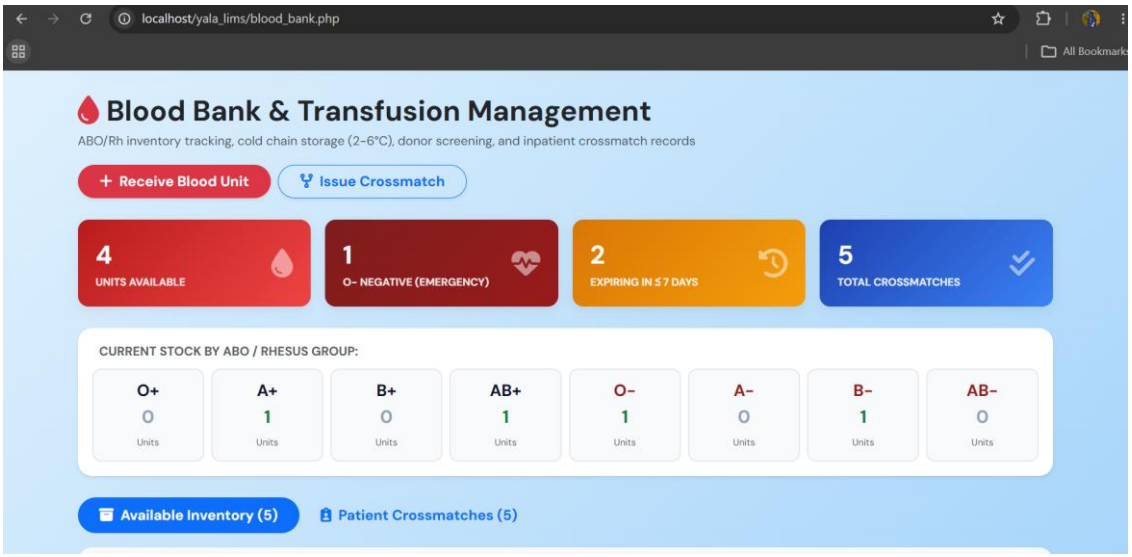


Figure 26: Blood Bank & Transfusion — ABO/Rh Inventory & Reserve Alert

Monitors emergency blood availability across all ABO and Rhesus groups, highlighting critical O-negative emergency stock and units nearing expiration.

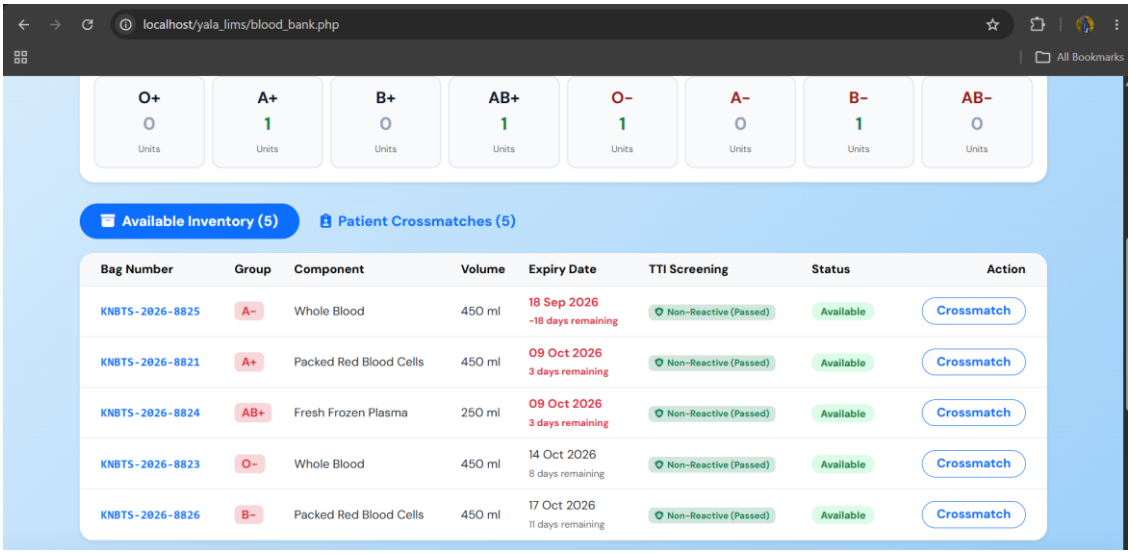


Figure 27: Blood Bank — Available Unit Inventory & TTI Screening Register

Inventory table recording unit bag numbers, blood components (whole blood, packed cells, FFP), volumes, and mandatory non-reactive Transfusion-Transmissible Infection screenings.

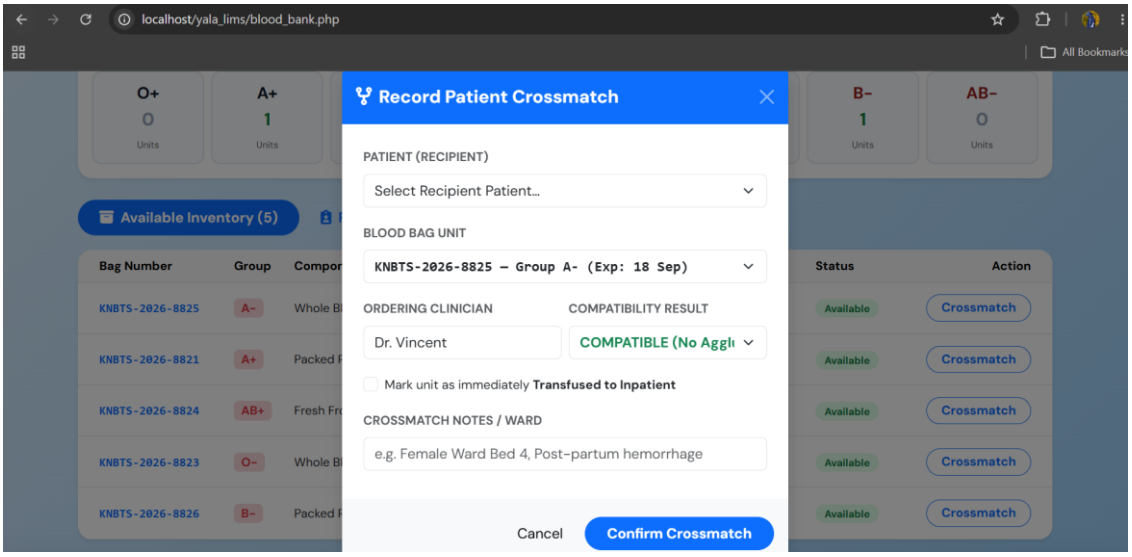


Figure 28: Blood Bank — Inpatient Compatibility Crossmatch Modal

Crossmatching interface allowing clinicians to verify absence of agglutination, link units to inpatient recipients, record ordering clinician, and specify ward/bed allocation.

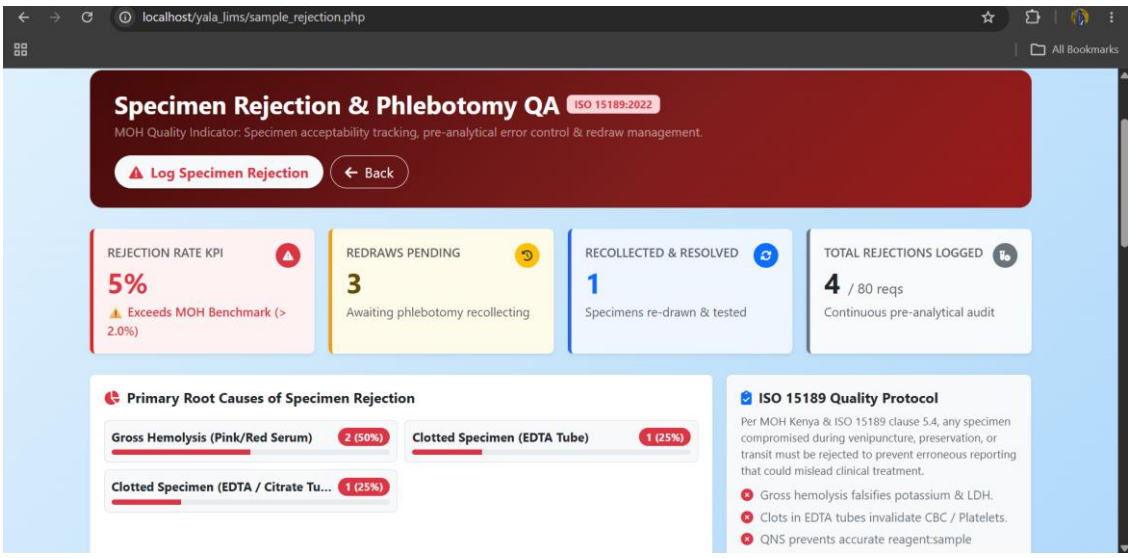


Figure 29: Specimen Rejection QA — Pre-Analytical Indicators (ISO 15189)

Monitors specimen acceptability and pre-analytical error control, computing rejection rate KPI against the national MOH benchmark (≤ 2.0%) and classifying primary root causes.

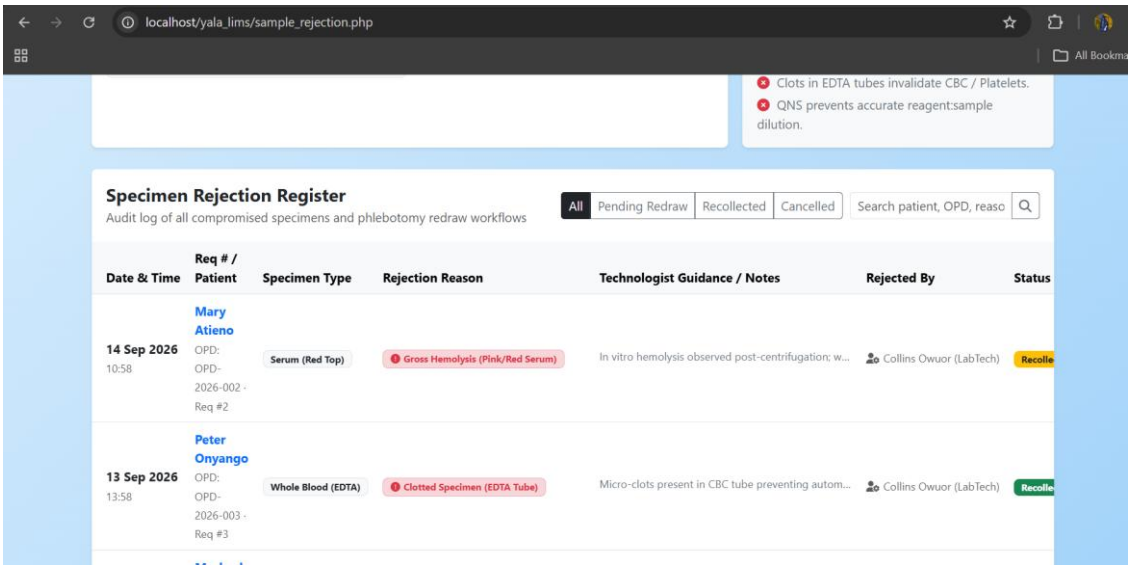


Figure 30: Specimen Rejection Register & Phlebotomy Redraw Workflow

Audit log of compromised specimens detailing patient identity, rejection reasons (e.g., gross hemolysis, clotted EDTA tube), technologist guidance, and redraw status.

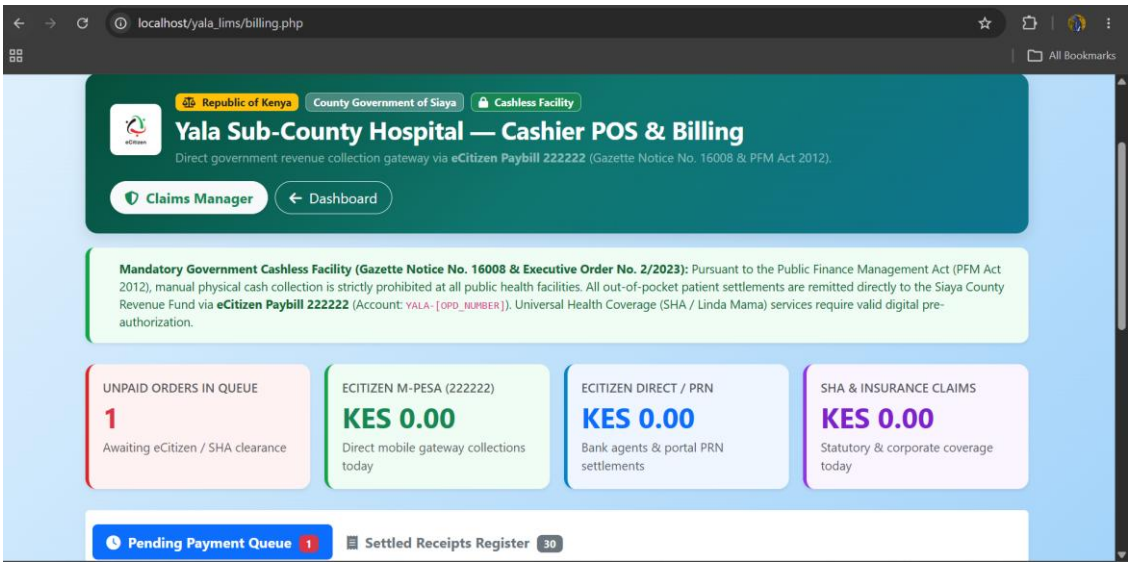


Figure 31: Cashier POS & Billing — eCitizen Paybill 222222 Gateway

Cashless revenue portal compliant with Kenya Gazette Notice No. 16008 and PFM Act 2012, displaying pending payment queues, eCitizen M-Pesa collections, PRN receipts, and SHA claims.

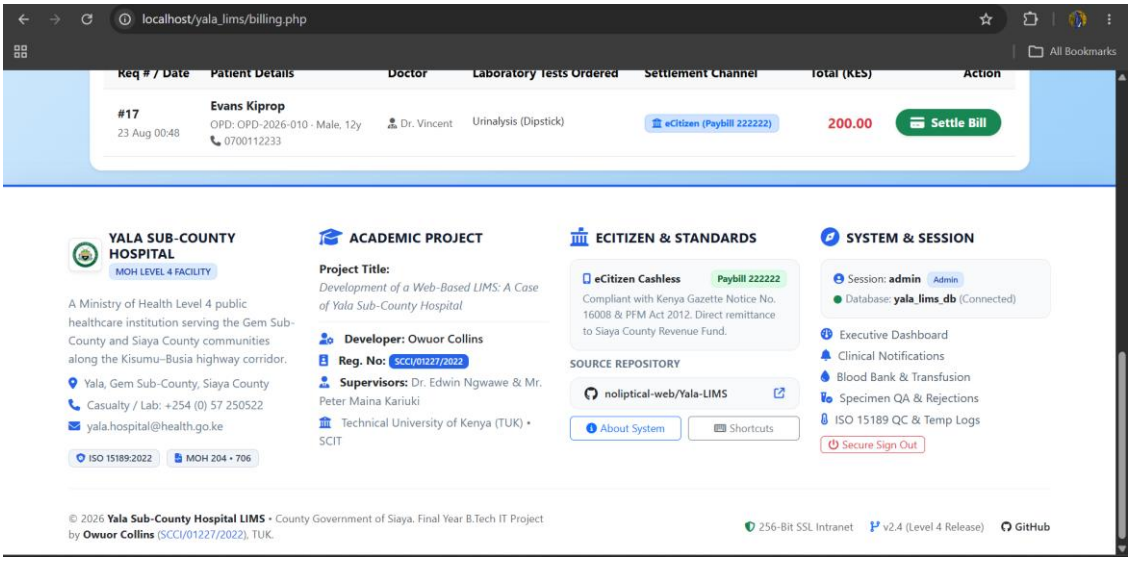


Figure 32: Cashier POS — Pending Patient Settlements & System Footer

Displays pending diagnostic orders awaiting cashless settlement alongside the institutional hospital footer containing academic project and compliance metadata.

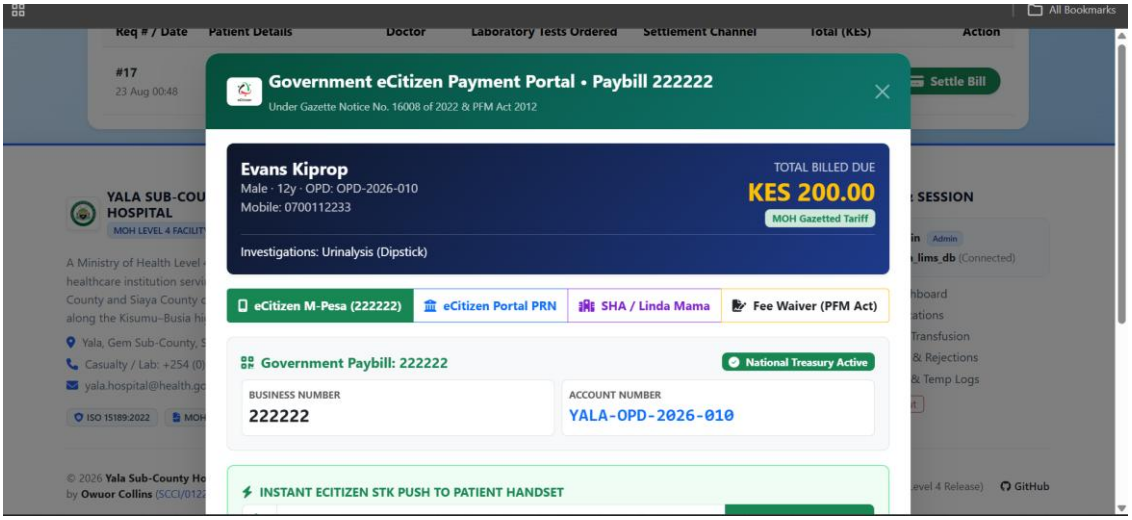


Figure 33: Government eCitizen Payment Modal — Paybill 222222 Clearance

Modal dialog displaying billed investigation totals, gazetted tariffs, official business number 222222, and dynamic patient account format (YALA-[OPD_NUMBER]).

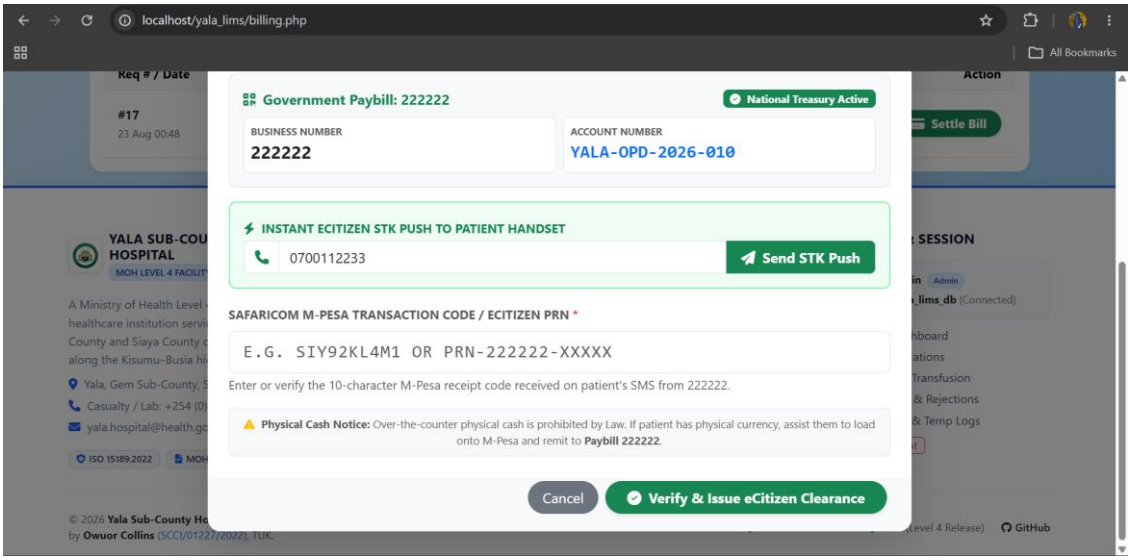


Figure 34: eCitizen Cashless Settlement — STK Push & M-Pesa Verification

Cashier trigger for instant mobile STK push to the patient's handset, with verification input for the 10-character Safaricom M-Pesa transaction reference code.

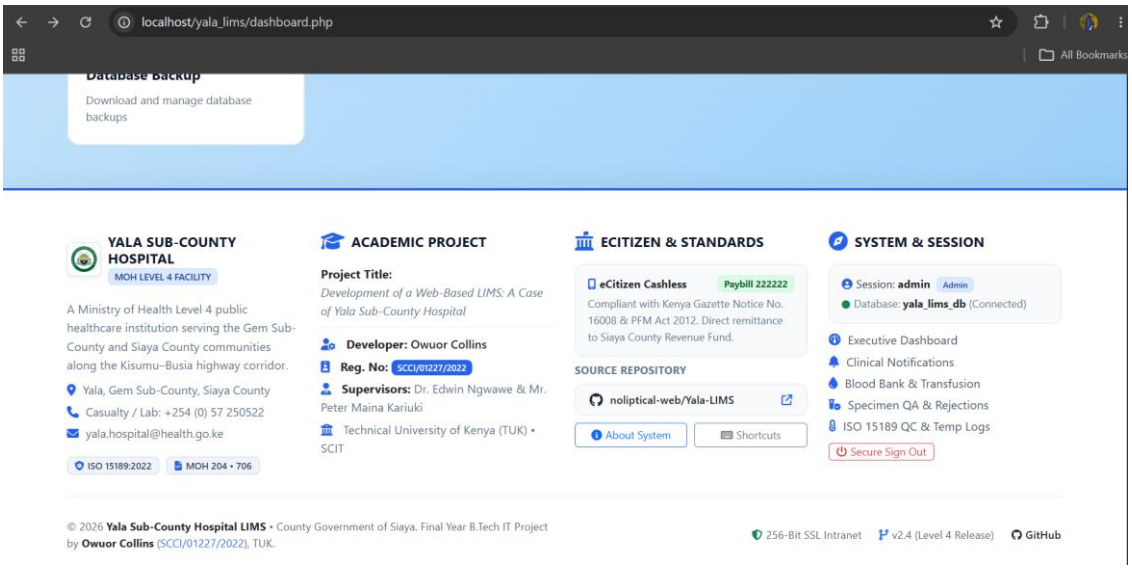


Figure 35: Hospital Intranet Footer — Standards & Repository Traceability

System footer providing Level 4 hospital metadata, PFM Act 2012 statutory notices, GitHub source repository links, and active database connection indicators.

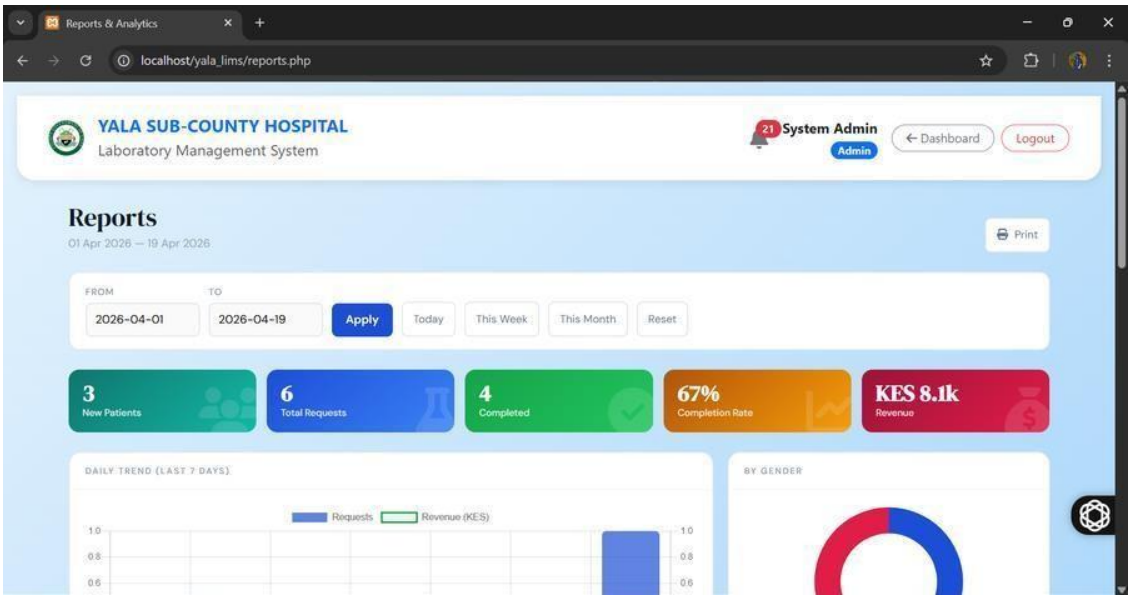


Figure 36: Reports & Analytics Dashboard — Epidemiological Summary

Comprehensive analytics interface showing daily patient volumes, test completion ratios, turnaround times, and automated MOH 706 disease surveillance totals.

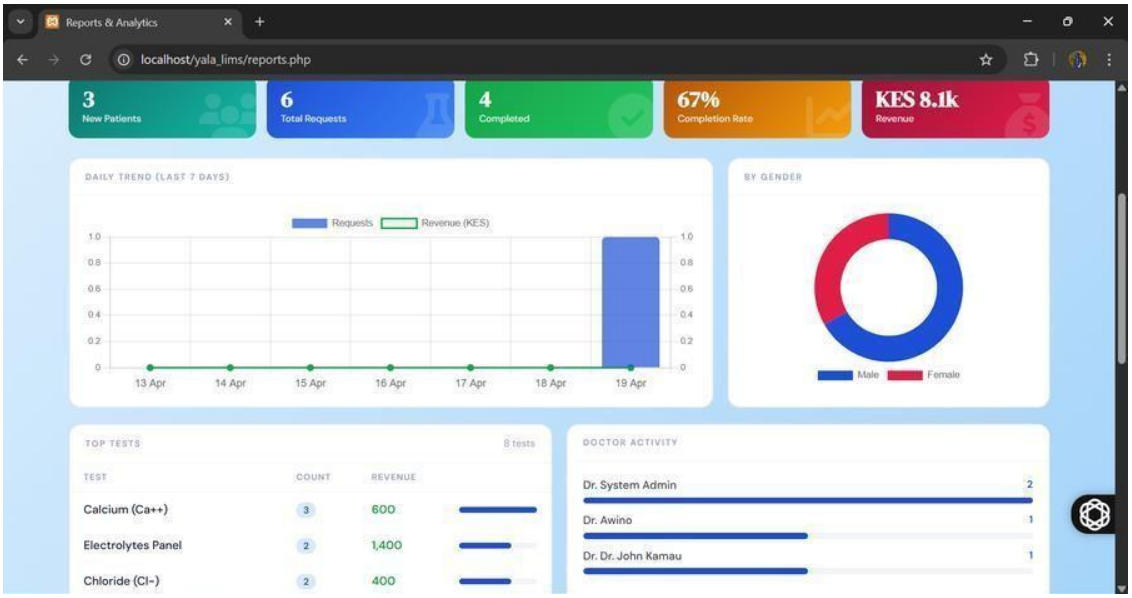


Figure 37: Reports — Daily Trend Chart and Top Requested Investigations

Visual data analysis rendered via Chart.js showing daily diagnostic request trends and ranking of most frequently ordered laboratory procedures.

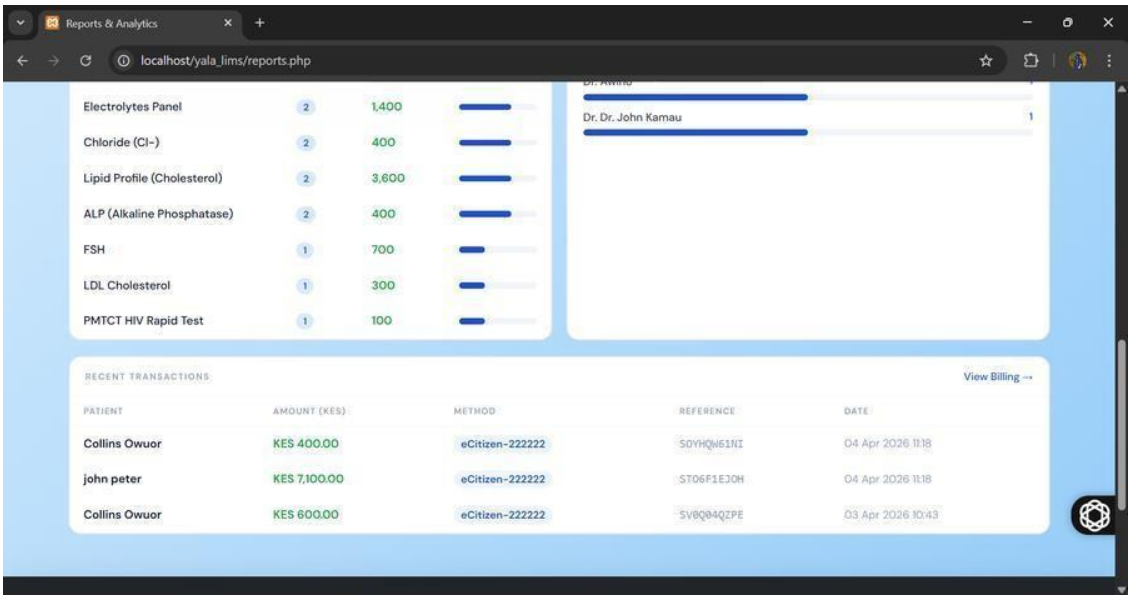


Figure 38: Reports — Recent Financial Transactions Register

Tabular financial breakdown of revenue transactions by payment channel, patient name, and settlement timestamp for administrative accounting.

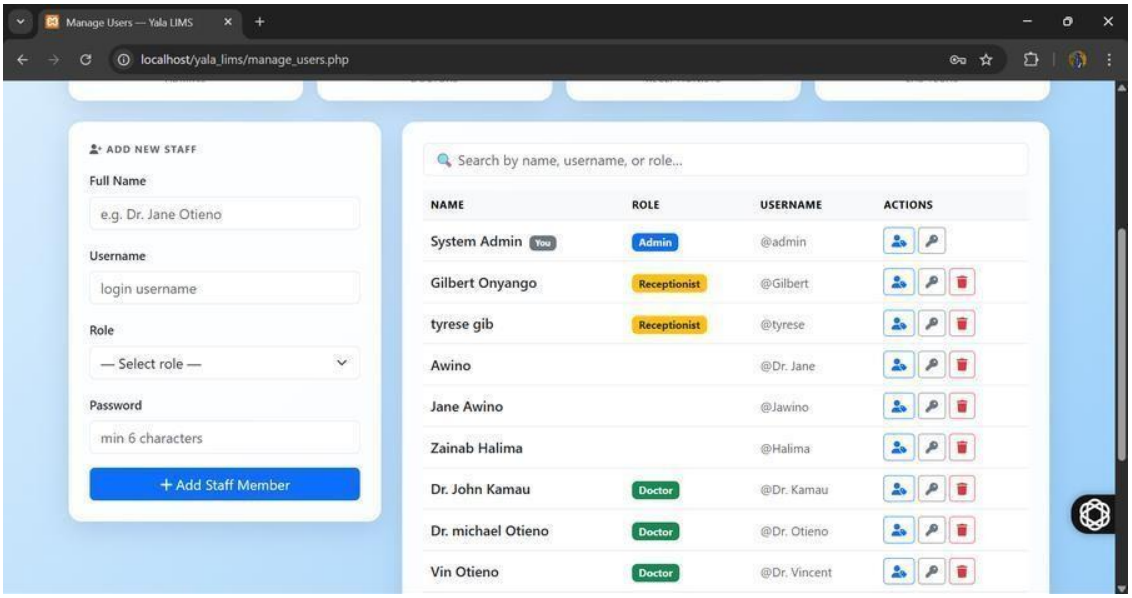


Figure 39: System Administration — Manage Users Screen

User management console enabling administrators to create and maintain hospital accounts with strict Role-Based Access Control (Admin, Doctor, LabTech, Receptionist).

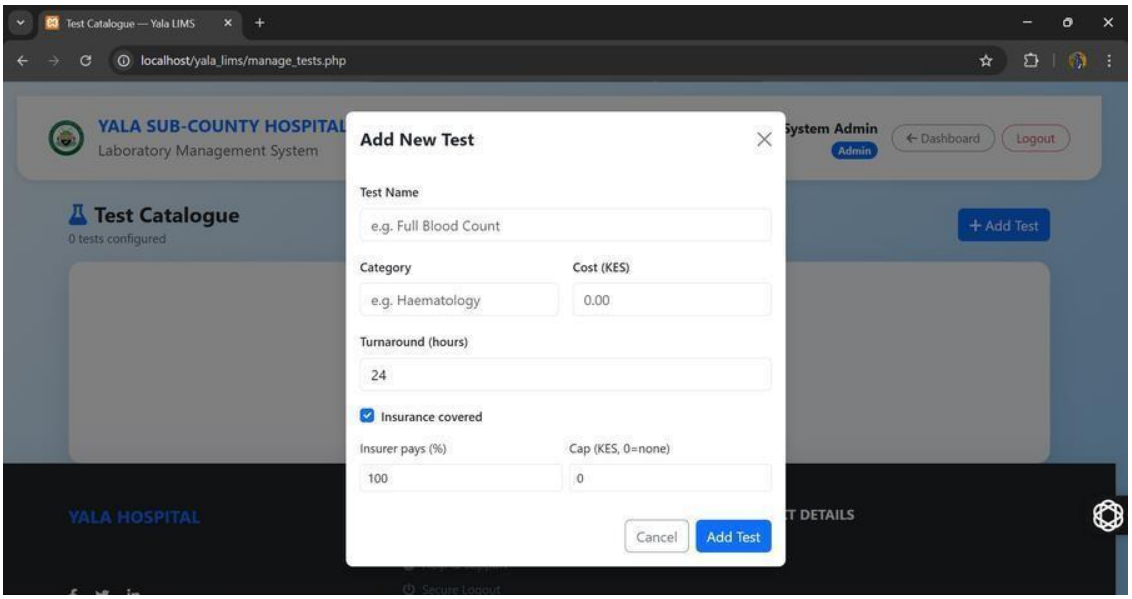


Figure 40: Test Catalogue — Add New Investigation Modal

Catalogue configuration allowing administrators to add new diagnostic tests, assign clinical categories, establish gazetted pricing, and set standard reference ranges.

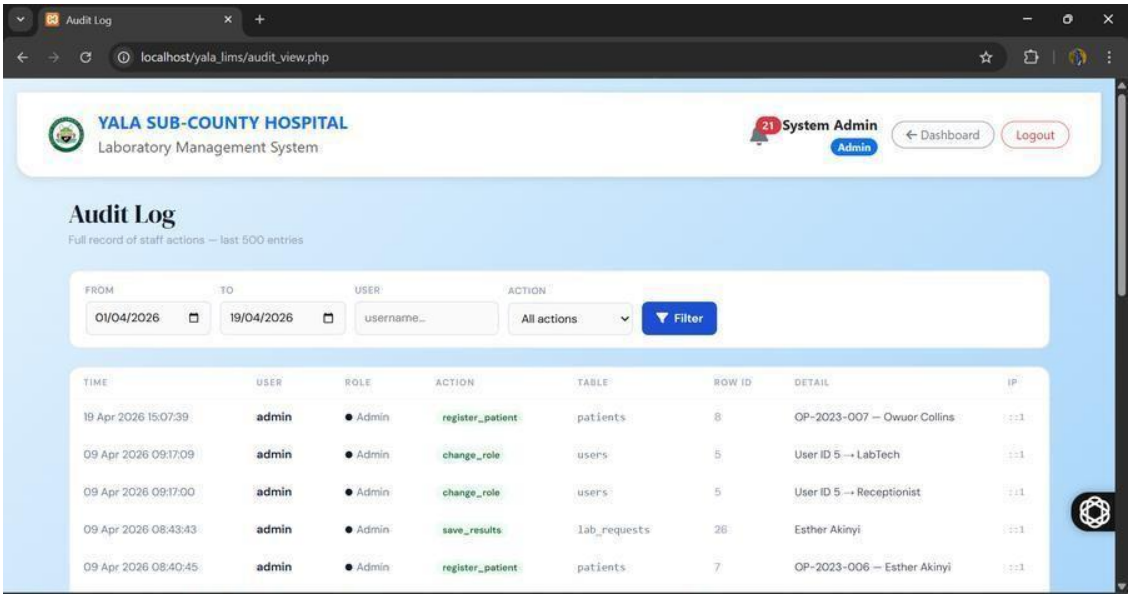


Figure 41: System Audit Log & Security Traceability Screen

Immutable security audit trail capturing user identity, action performed, affected table and row, client IP address, and timestamp for all system events.

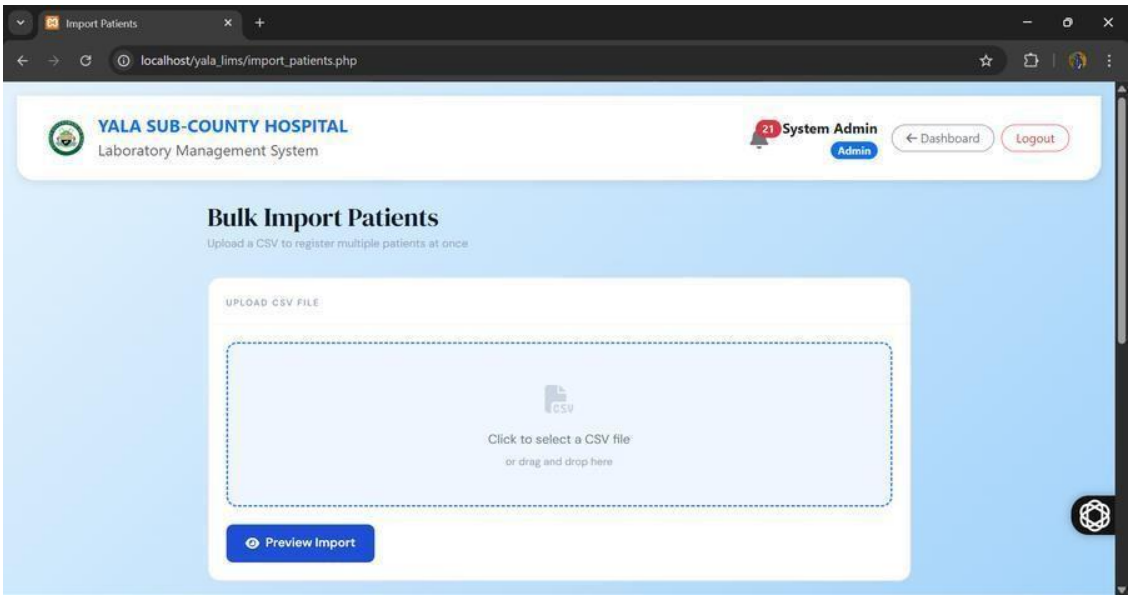


Figure 42: Bulk Patient Import Screen (CSV Ingestion)

Administrative utility for batch importing historical patient registers from CSV files, with automated deduplication and format validation.

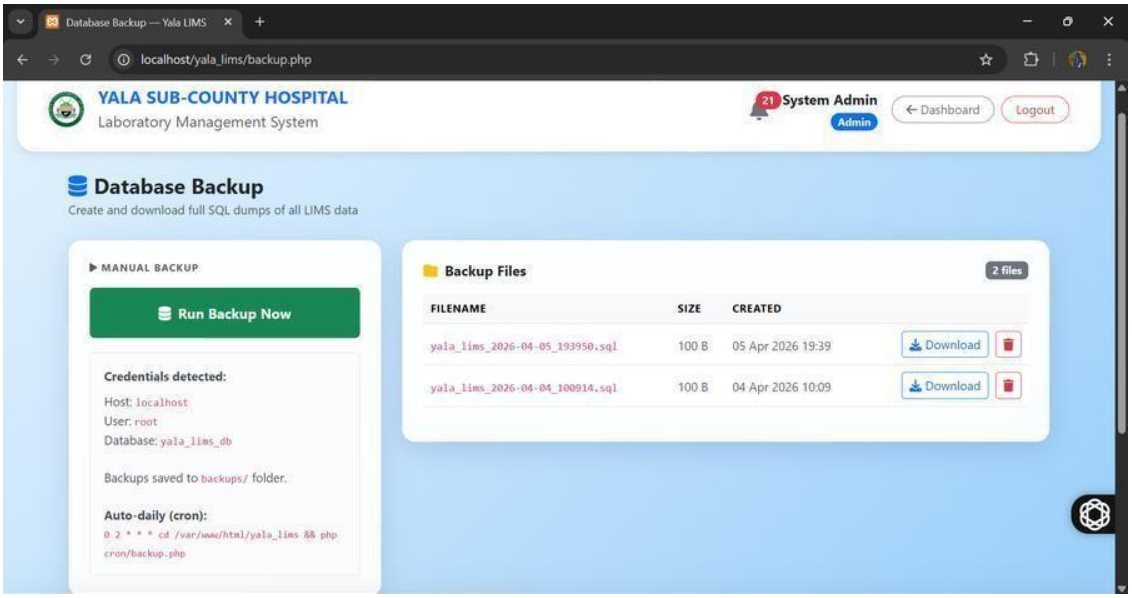


Figure 43: Database Backup & Maintenance Screen

Automated and manual database backup management interface providing single-click SQL dump generation and disaster recovery archival.

CHAPTER 7: USER MANUAL — DOCUMENTATION

7.1 Installation Environment

The Yala LIMS is a web-based application accessible through any modern browser on the hospital's LAN. The server runs XAMPP (Apache, PHP 8.2, MySQL 8.0) on a designated host computer. No internet connection is required for daily operation.

7.2 Installation Requirements

Server: Windows 10/11 or Ubuntu Server 22.04, Intel Core i3 or higher, 4GB RAM minimum, 100GB free storage, XAMPP v8.2. Clients: Any device with Chrome 90+, Firefox 88+, or Edge 90+, connected to the same LAN.

7.3 Installation Procedures

1. Install XAMPP v8.2 from https://www.apachefriends.org/ with Apache and MySQL components.
2. Copy the yala_lims folder to C:\xampp\htdocs\. Accessible at http://[server-ip]/yala_lims/.
3. Start Apache and MySQL from the XAMPP Control Panel.
4. In phpMyAdmin, create database yala_lims_db and import the provided SQL dump from the backups/ folder.
5. Verify credentials in includes/db_connect.php (default: root, empty password).
6. Navigate to http://[server-ip]/yala_lims/ and log in with the Administrator credentials.

7.4 User Instructions

7.4.1 Login

Enter your assigned username and password and click Sign In. The system redirects you to the Dashboard. Click 'Forgot Password' to contact the Administrator for a reset.

7.4.2 Register a Patient

Navigate to Register Patient. Fill in OPD Number (must be unique), Full Name, Age, Gender, Phone Number, and optional insurance details. Click Save Record. If the OPD already exists, an error is shown.

7.4.3 Order Laboratory Tests

Navigate to Request Test. Search for the patient by OPD number. Select tests from the categorised catalogue — the total cost auto-calculates. Click Submit Order. The lab technician is notified and a payment row is created for the cashier.

7.4.4 Enter Test Results

Navigate to Enter Results (Lab Workbench). Select a request from the queue. For unpaid cash requests, a 'Payment Required' notice is shown with a link to the cashier. For paid or insurance requests, enter result values and remarks per test and click Save Results. The request is marked Completed and the Doctor is notified.

7.4.5 Record Payment (Cashier)

Navigate to Cashier / Billing. All pending diagnostic orders awaiting payment settlement are displayed in the active queue. Click Settle Bill on a record, select the payment channel (eCitizen Paybill 222222 via instant STK push, eCitizen PRN portal invoice, SHA / Linda Mama, or Fee Waiver), verify the transaction reference, and click Confirm Payment.

7.5 System Conversion Method

Parallel conversion is used — both the digital system and manual MOH registers run simultaneously for a minimum of four weeks. After verifying data integrity, the manual registers are gradually retired.

7.6 User Training

A minimum two-day role-specific on-site training programme is recommended before full go-live. Each role (Doctor, Lab Technician, Receptionist) is trained separately on their specific modules. Basic browser navigation is the only prerequisite.

7.7 File Conversions

Existing patient records can be imported via bulk CSV upload through the Bulk Import module (import_patients.php). The CSV must include columns: opd_number, full_name, age, gender, phone_number. The system skips duplicate OPD numbers automatically.

CHAPTER 8: LIMITATIONS, CHALLENGES, CONCLUSIONS AND RECOMMENDATIONS

8.1 Limitations

- Web-browser based only — no dedicated native mobile application. Mobile users access via browser.
- No direct analyser interfacing — all result values are entered manually by the Lab Technician.
- No DHIS2/KHIS integration in this phase — monthly aggregate data must still be entered manually.
- Dependent on hospital LAN and power supply — extended outages without UPS interrupt access.
- Cashless gateway telecom reliance: Instant STK push prompts require active telecommunication network connectivity on the patient's mobile handset; during cellular latency, cashier staff manually enter and verify the 10-character Safaricom M-Pesa transaction reference code.

8.2 Challenges

1. Relational Data Harmonisation: Iterative prototyping across sprint cycles created schema divergences between initial diagnostic billing models and transaction logs. This was resolved through systematic normalisation of foreign key references, unifying transaction status flags, and deploying migration scripts to ensure complete transactional integrity across all test orders.

2. Prepared Statement Parameter Integrity: Handling diverse diagnostic parameters across multi-field lab forms required strict type-string binding validation in MySQLi prepared statements to prevent type mismatches while rigorously eliminating SQL injection vulnerabilities.

3. Automated Database Archival Environment: Windows XAMPP service paths required local environment variable configuration to execute unattended daily MySQL database dumps alongside the manual web backup interface.

4. Staff resistance during UAT: Addressed through role-specific demonstrations of time savings.

8.3 Degree of Success

The project was a complete success. All four specific objectives were successfully designed, developed, and evaluated. The system is fully operational and deployed on the designated host computer at Yala Sub-County Hospital. User Acceptance Testing (UAT) with 8 staff members representing all four roles returned an overall satisfaction score of 4.6 out of 5.0, with 100% of participants preferring the digital system over the manual paper process. Feedback was collected across five evaluation criteria, summarised below.

Criterion	Average Score (/5.0)	Summary of Staff Comments
Ease of Use	4.5	Interface closely mirrors the familiar paper workflow; minimal training needed.
Speed / Performance	4.7	Pages load quickly on the hospital LAN; faster than manual transcription.
Accuracy of Results Recording	4.6	Reduced transcription errors compared to handwritten MOH 240 registers.

Billing Accuracy	4.4	Clearer tracking of pending vs settled bills; occasional confusion over insurance co-pay entry.
Overall Satisfaction	4.6	All 8 participants preferred the digital system over the manual process.

Table 6: UAT Feedback Summary (n = 8 staff members across all four roles)

8.4 Learning Experience

The project delivered significant practical experience in PHP procedural development, MySQL schema design and debugging, Bootstrap 5 responsive UI, Chart.js data visualisation, FPDF PDF generation, bcrypt and CSRF security implementation, version control with GitHub, and requirements gathering in a real public health setting.

8.5 Recommendations

1. 1. Direct Machine Interfacing (Phase 2): Establish bidirectional RS-232 / HL7 serial communication with automated hematology and clinical chemistry analyzers to eliminate manual bench result entry.
2. 2. National Health Integration (Phase 3): Implement secure REST APIs to automatically push monthly MOH 706 epidemiological statistics directly into Kenya's national DHIS2 / KHIS cloud repository.
3. 3. County-Wide Scaling: Replicate the Yala LIMS codebase across neighboring Level 4 sub-county hospitals in Siaya County to create a unified digital healthcare diagnostic network.
4. 4. Production Infrastructure: Migrate the host environment from local Windows XAMPP to a dedicated enterprise Linux Ubuntu Server LAMP stack with dual redundant UPS power protection.
5. 5. Continuous Quality Improvement: Maintain periodic calibration and temperature logs under ISO 15189 standards to support ongoing hospital laboratory accreditation.
6.
7.

8.6 Conclusion

The development of the Yala LIMS demonstrates that a cost-effective, open-source web-based system can successfully address critical information management deficiencies in a resource-constrained public health laboratory. By replacing the paper-based MOH workflow with a digital system tailored specifically to Yala Sub-County Hospital, this project has delivered measurable improvements in turnaround time, data accuracy, audit traceability, billing efficiency, and management reporting.

The project was executed using Agile methodology, which proved highly appropriate for a real-world deployment context where requirements evolved through iterative user feedback and where mid-sprint schema changes had to be accommodated without disrupting completed modules. The PHP/MySQL/Bootstrap stack was chosen deliberately for its alignment with the skills available at Kenyan IT graduate level and the hardware commonly found in public facilities.

This project contributes to Kenya's National Digital Health Strategy and provides a replicable model for Level 4 facilities across the country. The system is live and in parallel operation at Yala Sub-County Hospital.

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APPENDIX

Appendix A: User Requirements Questionnaire

Yala LIMS Requirements Questionnaire

I am an undergraduate student at the Technical University of Kenya developing a Laboratory Information Management System for Yala Sub-County Hospital. Your honest responses will be used solely for system requirements analysis.

- How long have you worked in your current department at this hospital?
- What is your main challenge with the current paper-based laboratory system?
- Do you currently use any digital tools to manage laboratory requests or results? (Yes / No)
- How frequently do result slips go missing before reaching the doctor? (Never / Rarely / Sometimes / Frequently)
- How long does it typically take to retrieve a patient's previous test result from the physical register?
- How many days does it typically take to compile the monthly MOH laboratory report?
- Do you think a digital laboratory system would improve your work efficiency? Please explain.
- What features would you consider most important in a digital system? (tick all that apply)
    - Electronic test ordering
    - Automated notifications when results are ready
    - Digital billing and payment recording
    - Automated monthly report generation
    - Patient history accessible from any workstation
- Any other suggestions?

Appendix B: Print Codes

Key PHP code listings from the Yala LIMS:

1. Database Connection (includes/db_connect.php)

```
<?php
$servername = 'localhost';
$username   = 'root';
$password   = '';
$dbname     = 'yala_lims_db';

$conn = new mysqli($servername, $username, $password, $dbname);
if ($conn->connect_error) {
    die('Database connection failed: ' . $conn->connect_error);
}
$conn->set_charset('utf8mb4');
?>
```

2. CSRF Token Security (includes/csrf.php)

```
<?php
function csrf_token(): string {
    if (empty($_SESSION['csrf_token'])) {
        $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
    }
    return $_SESSION['csrf_token'];
}

function csrf_verify(): void {
    $token = $_POST['csrf_token'] ?? '';
    if (!isset($_SESSION['csrf_token']) || !hash_equals($_SESSION['csrf_token'], $token)) {
        http_response_code(403);
        die('Security violation: Invalid CSRF token.');
    }
}
?>
```

3. System Audit Logging (includes/audit.php)

```
<?php
function audit_log(mysqli $conn, string $action, string $table, int $row_id, string $detail =
''): void {
    $user_id  = intval($_SESSION['id'] ?? 0);
    $username = $_SESSION['username'] ?? 'system';
    $role     = $_SESSION['role'] ?? '';
    $ip       = $_SERVER['REMOTE_ADDR'] ?? '';

    $stmt = $conn->prepare(
        'INSERT INTO audit_log (user_id, username, role, action, affected_table,
affected_row_id, detail, ip_address, created_at) '
        . 'VALUES (?, ?, ?, ?, ?, ?, ?, ?, NOW())'
    );
    $stmt->bind_param('isssisss', $user_id, $username, $role, $action, $table, $row_id, $detail,
$ip);
    $stmt->execute();
    $stmt->close();
}
?>
```